0.1 Overall Heat-Transfer Paths and Coupled Energy Balances

The MATLAB model represents the heater and vermicomposter as a set of coupled heat-transfer paths rather than a single series resistance chain. The principal paths are:

1. Electrical heating in the nichrome wire, with convection from the wire to the heater-tube airflow.
2. Loss from the heater-tube airflow to ambient through the heater tube wall and insulation.
3. Useful heat transfer from the aeration airflow to the substrate by
   - transfer through the aeration tube wall,
   - direct warm-air release into the bed through the perforations, and
   - subtraction of a latent heat sink associated with evaporation from the wetted substrate.
4. Heat loss from the bed to ambient through the vessel wall.

0.1.1 Wire-to-Air Convection

The wire-to-air convective coefficient is computed as

$$h_{\text{wire}} = \frac{Nu_{\text{wire}}\, k}{D_{\text{wire}}} \tag{1}$$

where Nu_{wire} is obtained from the Churchill–Bernstein cylinder-in-crossflow correlation [R1]:

$$Nu_{\text{wire}} = 0.3 + \frac{0.62\, Re_{\text{wire}}^{1/2} Pr^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{Re_{\text{wire}}}{282000}\right)^{5/8}\right]^{4/5}. \tag{2}$$

Equivalently, matching the code implementation,

$$Nu_{\text{wire}} = 0.3 + \left(\frac{0.62\, Re_{\text{wire}}^{1/2} Pr^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}}\right) \left[1 + \left(\frac{Re_{\text{wire}}}{282000}\right)^{5/8}\right]^{4/5}. \tag{3}$$

The same Churchill–Bernstein form is also used for the perforation hole-scale crossflow convection in the moisture model.

This coefficient is used to estimate the local wire-to-air temperature rise:

$$\Delta T_{\text{wire}} = \frac{q_{\text{gen}}}{h_{\text{wire}} A_{\text{wire,local}}} \tag{4}$$

so that

$$T_{\text{wire}} = T_{\text{air}} + \Delta T_{\text{wire}}. \tag{5}$$

This term affects wire temperature prediction, but it is not part of an overall U value.

0.1.2 Heater-Tube Loss to Ambient

The internal convection coefficient for the heater tube is

$$h_{\text{inside,heater}} = \frac{Nu_{\text{tube}}\, k}{D_i}. \tag{6}$$

The implemented internal-tube Nusselt correlation is piecewise. For laminar developing flow the code uses a Hausen-type entrance-length correction [R5]:

$$Gz = Re\, Pr\, \frac{D_i}{L}, \tag{7}$$

$$Nu_{\text{tube}} = 3.66 + \frac{0.0668\, Gz}{1 + 0.04\, Gz^{2/3}}, \qquad Re < 2300. \tag{8}$$

For turbulent flow the code uses Gnielinski [R2]:

$$Nu_{\text{tube}} = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}, \qquad Re \geq 2300, \tag{9}$$

with the lower bound

$$Nu_{\text{tube}} \geq 3.66. \tag{10}$$

Here f is the Churchill friction factor [R3] used consistently in both the pressure-drop and turbulent heat-transfer calculations.

The overall heat-transfer coefficient from the heater-tube air to ambient is modeled on the exposed outer-area basis. With the insulated exposed diameter

$$OD_{\text{ext}} = OD + 2t_{\text{ins}}, \tag{11}$$

this gives

$$U_{\text{heater}} = \left(\frac{OD_{\text{ext}}}{ID\, h_{\text{inside,heater}}} + \frac{OD_{\text{ext}}}{2k_{\text{wall}}} \ln\left(\frac{OD}{ID}\right) + \frac{OD_{\text{ext}}}{2k_{\text{ins}}} \ln\left(\frac{OD_{\text{ext}}}{OD}\right) + \frac{1}{h_{\text{outside,heater}}} \right)^{-1}. \tag{12}$$

The corresponding heat loss from the heater section is

$$q_{\text{loss,heater}} = U_{\text{heater}}\, P_{\text{out}}\, \Delta x\, \left(T_{\text{air}} - T_{\text{env}}\right), \qquad P_{\text{out}} = \pi OD_{\text{ext}}. \tag{13}$$

This loss is coupled to the heater air-energy balance through

$$q_{\text{to air}} = q_{\text{gen}} - q_{\text{loss,heater}} \tag{14}$$

and

$$\dot{m} c_p\, dT = q_{\text{to air}}. \tag{15}$$

Thus, U_{heater} determines how much electrical power remains in the airflow versus being lost directly to ambient.

0.1.3 Aeration-Tube Heat Transfer to the Bed

The internal aeration-tube convection coefficient is

$$h_{\text{inside,aer}} = \frac{Nu_{\text{tube}}\, k}{D_i} \tag{16}$$

and the external bed-side coefficient is taken as

$$h_{\text{outside,aer}} = h_{\text{bed}}. \tag{17}$$

The same internal-tube Nu_{tube} branch logic given above is used for the aeration tube as for the heater tube: Hausen-type laminar developing flow for $Re < 2300$ and Gnielinski for $Re \geq 2300$.

The overall coefficient from aeration air to bed through the aeration tube wall is

$$U_{\text{aer}} = \left(\frac{OD}{ID\, h_{\text{inside,aer}}} + \frac{OD}{2k_{\text{wall}}} \ln\!\left(\frac{OD}{ID}\right) + \frac{1}{h_{\text{outside,aer}}} \right)^{-1}. \tag{18}$$

Wall-mediated heat transfer from air to bed is then

$$q_{\text{wall}} = U_{\text{aer}}\, P_{\text{out}}\, \Delta x\, \left(T_{\text{air}} - T_{\text{bed}}\right). \tag{19}$$

The model also includes direct jet heating from released warm air:

$$q_{\text{jet}} = \dot{m}_{\text{release}}\, c_p\, \left(T_{\text{air}} - T_{\text{bed}}\right). \tag{20}$$

In the current implementation, evaporation from the wetted substrate is modeled explicitly as a latent sink:

$$q_{\text{evap}} = \dot{m}_{\text{evap}}\, h_{fg}. \tag{21}$$

Hence the branch heat delivered to the bed is

$$Q_{\text{bed,branch}} = \sum \left(q_{\text{wall}} + q_{\text{jet}} - q_{\text{evap}}\right) \tag{22}$$

and the total heat delivered by all parallel tubes is

$$Q_{\text{bed,total}} = N_{\text{tubes}}\, Q_{\text{bed,branch}}. \tag{23}$$

Therefore, U_{aer} sets the wall-transfer contribution to useful bed heating, q_{jet} adds a parallel direct-heating path, and q_{evap} reduces the net useful heat delivered to the substrate.

The bed-side coefficient h_{bed} remains a *sensible* wall-transfer coefficient. Latent evaporation is modeled separately through q_{evap} and is not absorbed into h_{bed}.

0.1.4 Vessel-Wall Heat Loss to Ambient

For the bin exterior, the natural-convection coefficient is obtained from

$$Ra = \frac{g\beta\Delta T L^3}{\nu\alpha}, \tag{24}$$

followed by the Churchill–Chu vertical-plate relation [R4],

$$Nu = \left[0.825 + \frac{0.387\, Ra^{1/6}}{\left(1 + (0.492/Pr)^{9/16}\right)^{8/27}}\right]^2, \tag{25}$$

and then

$$h_{\text{ext}} = \frac{Nu\, k}{L}. \tag{26}$$

The overall vessel-wall coefficient is

$$U_{\text{wall}} = \left(\frac{1}{h_{\text{inside,bin}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{ext}}}\right)^{-1}. \tag{27}$$

The total bed-to-ambient heat loss is

$$Q_{\text{loss}} = U_{\text{wall}} A_{\text{total}} \left(T_{\text{bin}} - T_\infty\right). \tag{28}$$

The same U_{wall} is used to construct the nodewise ambient-loss terms:

$$UA_{\text{bottom}} = U_{\text{wall}} A_{\text{bottom,node}}, \qquad UA_{\text{top}} = U_{\text{wall}} A_{\text{top,node}}. \tag{29}$$

Thus, U_{wall} governs the final loss of stored bed heat to ambient.

0.1.5 Coupling Through the Bed Energy Balances

The bed is represented as two lumped thermal nodes: a bottom node and a top node. Heat is added to these nodes from the aeration system, exchanged internally between the two bed regions, and lost to ambient through the vessel boundaries.

Heat input split between bottom and top nodes The total useful heat delivered from the aeration system is

$$Q_{\text{bed,total}} = N_{\text{tubes}} Q_{\text{bed,branch}}. \tag{30}$$

This heat is then partitioned between the bottom and top bed nodes using the parameter f_b:

$$Q_b = f_b\, Q_{\text{bed,total}}, \qquad Q_t = (1 - f_b)\, Q_{\text{bed,total}}. \tag{31}$$

Here, f_b is an assumed heat-input split parameter that represents the heater tube proximity to the bottom of the bed. It does *not* represent ambient heat loss.

Internal coupling between bottom and top bed regions The two nodes are thermally coupled through an internal conduction-like term:

$$UA_{\text{internal}} = k_{\text{bed}} \frac{A_{\text{internal}}}{\delta_{\text{internal}}}, \tag{32}$$

where

$$A_{\text{internal}} = LW \tag{33}$$

is the horizontal interface area between the lower and upper bed regions, and

$$\delta_{\text{internal}} \approx \frac{H}{2} \tag{34}$$

is the assumed characteristic distance between the node centroids.

This term allows heat to move between the bottom and top nodes:

$$Q_{\text{internal}} = UA_{\text{internal}}(T_t - T_b). \tag{35}$$

Ambient heat loss areas for each node The bottom node loses heat through the bottom surface and half of the side-wall area:

$$A_{\text{bottom,node}} = A_{\text{bottom}} + \frac{1}{2} A_{\text{side}}, \tag{36}$$

where

$$A_{\text{bottom}} = LW \tag{37}$$

and

$$A_{\text{side}} = 2HL + 2HW. \tag{38}$$

Thus the bottom-node ambient loss coefficient is

$$UA_{\text{bottom}} = U_{\text{wall}} A_{\text{bottom,node}}. \tag{39}$$

For the top node, the treatment depends on whether the bin top is covered or open.

If the top is treated as covered,

$$UA_{\text{top}} = U_{\text{wall}} \left(A_{\text{top}} + \frac{1}{2} A_{\text{side}} \right), \tag{40}$$

where

$$A_{\text{top}} = LW. \tag{41}$$

If the top is treated as open,

$$UA_{\text{top}} = U_{\text{wall}} \left(\frac{1}{2} A_{\text{side}} \right) + h_{\text{top}} A_{\text{top}}. \tag{42}$$

Therefore, with an open top:

- side-wall losses are still present and are split equally between the two nodes, and
- the top surface itself loses heat directly to ambient by convection rather than through the insulated wall stack.

Steady two-node balance Under steady conditions, the node temperatures T_b and T_t satisfy

$$(UA_{\text{bottom}} + UA_{\text{internal}})T_b - UA_{\text{internal}}T_t = Q_b + UA_{\text{bottom}}T_{\text{amb}} \tag{43}$$

and

$$-UA_{\text{internal}}T_b + (UA_{\text{top}} + UA_{\text{internal}})T_t = Q_t + UA_{\text{top}}T_{\text{amb}}. \tag{44}$$

These equations show that each node is heated by its assigned fraction of the aeration heat input, cooled by ambient loss through its assigned external area, and coupled to the other node through internal heat exchange.

Transient two-node balance In the transient model, the node temperatures evolve according to

$$\frac{dT_b}{dt} = \frac{Q_b + UA_{\text{internal}}(T_t - T_b) - UA_{\text{bottom}}(T_b - T_{\text{amb}})}{C_b} \tag{45}$$

and

$$\frac{dT_t}{dt} = \frac{Q_t + UA_{\text{internal}}(T_b - T_t) - UA_{\text{top}}(T_t - T_{\text{amb}})}{C_t}. \tag{46}$$

Here, C_b and C_t are the thermal capacitances of the bottom and top nodes, respectively.

Interpretation The two-node bed model therefore contains three distinct mechanisms:

1. **Heater input distribution:** f_b determines how the delivered aeration heat is partitioned between the bottom and top nodes.
2. **Internal redistribution:** UA_{internal} allows heat to move between the two bed regions.
3. **Ambient loss:** UA_{bottom} and UA_{top} determine how each node loses heat to the surroundings through the bottom, side walls, and top surface.

This means that the top node may cool faster than the bottom node even if it receives less direct heater input, particularly when the top is modeled as open to ambient.

0.1.6 Summary of Coupling

The overall coupling structure is therefore:

$$\text{wire power} \rightarrow \text{heater air} \rightarrow \text{bed heat input} \rightarrow \text{bed storage} \rightarrow \text{ambient loss}. \tag{47}$$

More specifically:

- h_{wire} determines wire temperature relative to the heater air.
- U_{heater} determines how much heater-air energy is lost directly to ambient.
- U_{aer} determines how much aeration-air energy crosses the aeration tube wall into the bed.
- $Q_{\text{bed,total}}$ combines wall transfer, jet heating, and evaporative cooling into the bed-energy input.
- U_{wall} determines how quickly the bed loses heat through the vessel to ambient.

0.1.7 Meaning and Calculation of T_{air} and P_{out}

In the model, T_{air} denotes the temperature of the flowing air stream inside the heater tube or aeration tube, depending on the section being analyzed. It is not a fixed temperature; rather, it is updated segment-by-segment from an energy balance.

Heater-tube air temperature At the inlet to the heater tube, the air is assumed to enter at ambient greenhouse temperature:

$$T_{\text{air,in}} = T_{\text{env}}. \tag{48}$$

As the air moves through the heater, it gains energy from the electrically heated wire and loses some energy to the surroundings through the heater tube wall and insulation. For each axial segment, the net heat transferred to the air is

$$q_{\text{to air}} = q_{\text{gen}} - q_{\text{loss,heater}}, \tag{49}$$

where

$$q_{\text{gen}} = \frac{P_{\text{branch}}}{L}\,\Delta x \tag{50}$$

is the local rate of electrical heat generation assigned to the segment, and

$$q_{\text{loss,heater}} = U_{\text{heater}}\,P_{\text{out}}\,\Delta x\,(T_{\text{air}} - T_{\text{env}}) \tag{51}$$

is the local heat loss from the heater air to ambient.

The air temperature is then advanced according to

$$T_{\text{air,new}} = T_{\text{air,old}} + \frac{q_{\text{to air}}}{\dot{m}c_p}. \tag{52}$$

Thus, in the heater tube, T_{air} is the local heated-air temperature resulting from the balance between wire heating and heat loss to the surroundings.

Aeration-tube air temperature The air entering the aeration tube is the outlet air from the heater tube:

$$T_{\text{air,in,aer}} = T_{\text{air,out,heater}}. \tag{53}$$

As the air flows through the aeration tube, it transfers heat to the substrate by three coupled mechanisms:

1. wall-mediated transfer through the aeration tube wall,
2. direct sensible heat release with the discharged warm air, and
3. latent heat consumption by evaporation from the wetted substrate.

The wall-transfer term is

$$q_{\text{wall}} = U_{\text{aer}}\, P_{\text{out}}\, \Delta x\, \left(T_{\text{air}} - T_{\text{bed}}\right), \tag{54}$$

the direct jet term is

$$q_{\text{jet}} = \dot{m}_{\text{release}}\, c_p\, \left(T_{\text{air}} - T_{\text{bed}}\right), \tag{55}$$

and the evaporation term is

$$q_{\text{evap}} = \dot{m}_{\text{evap}}\, h_{fg}. \tag{56}$$

In the present implementation, the air temperature along the aeration tube is updated using the wall-transfer term:

$$T_{\text{air,new}} = T_{\text{air,old}} - \frac{q_{\text{wall}}}{\dot{m}_{\text{remaining}} c_p}. \tag{57}$$

Therefore, in the aeration section, T_{air} is the local air temperature after heater exit, reduced progressively as sensible heat is transferred from the air stream to the bed. The net bed heat delivered at the same segment is then corrected by subtracting the latent evaporation sink.

Meaning of P_{out} The quantity P_{out} is the *outside perimeter* of the cylindrical tube used to convert an axial segment length into an external surface area. For a tube of outside diameter D_{out},

$$P_{\text{out}} = \pi D_{\text{out}}. \tag{58}$$

Hence,

$$P_{\text{out}}\, \Delta x \tag{59}$$

is the outside surface area of one tube segment of length Δx.

This is why the wall heat-transfer terms are written in the form

$$q = U\, P_{\text{out}}\, \Delta x\, \Delta T : \tag{60}$$

the product $P_{\text{out}} \Delta x$ supplies the heat-transfer area, and $U \Delta T$ supplies the heat flux per unit area.

Summary In summary:

- T_{air} is the local temperature of the moving air stream.
- In the heater tube, it increases due to electrical heating minus ambient losses.
- In the aeration tube, it decreases as sensible heat is transferred to the substrate.
- The net useful bed heating in the aeration tube is further reduced by the latent evaporation term q_{evap}.
- P_{out} is the tube outside perimeter, so that $P_{\text{out}}\Delta x$ is the external area of each tube segment.

h_{top} for an uncovered top surface When the bin top is modeled as open to ambient, the top-surface heat loss is represented by a natural-convection coefficient h_{top}. This coefficient describes free convection between the exposed upper bed surface and the surrounding air.

It is defined by

$$h_{\text{top}} = \frac{Nu\,k}{L_c}, \tag{61}$$

where the characteristic length is taken as

$$L_c = \frac{A_{\text{top}}}{P_{\text{top}}}, \tag{62}$$

with

$$A_{\text{top}} = LW, \qquad P_{\text{top}} = 2(L + W). \tag{63}$$

The Rayleigh number for the open top is

$$Ra = \frac{g\beta\Delta T L_c^3}{\nu\alpha}. \tag{64}$$

Using the standard McAdams-style correlations for free convection from horizontal plates:

For a *heated upward-facing surface* (or equivalently a cooled downward-facing surface),

$$Nu = 0.54\,Ra^{1/4}, \qquad Ra \leq 10^7, \tag{65}$$

and

$$Nu = 0.15\,Ra^{1/3}, \qquad Ra > 10^7. \tag{66}$$

For a *heated downward-facing surface* (or equivalently a cooled upward-facing surface),

$$Nu = 0.27\,Ra^{1/4}, \tag{67}$$

Since the uncovered compost surface is warmer than the ambient air during heating operation, the physically relevant case is the heated upward-facing surface. Therefore, the top-node ambient heat loss is modeled as

$$Q_{\text{top,ambient}} = h_{\text{top}} A_{\text{top}} (T_t - T_{\text{amb}}). \tag{68}$$

In the two-node bed model, this contributes to the top-node ambient-loss coefficient as

$$U A_{\text{top}} = U_{\text{wall}} \left(\frac{1}{2} A_{\text{side}} \right) + h_{\text{top}} A_{\text{top}}, \qquad \text{for an open top.} \tag{69}$$

Thus, when the top is uncovered, the top-node heat loss consists of:

- side-wall loss through the vessel wall, represented by

$$U_{\text{wall}} \left(\frac{1}{2} A_{\text{side}} \right),$$

  and

- direct natural-convection loss from the exposed top surface, represented by

$$h_{\text{top}} A_{\text{top}}.$$

If the top is instead treated as covered, the model does not use h_{top}; it applies the covered-wall coefficient U_{wall} over the top area as well.

Total vessel heat loss The total vessel heat loss is the sum of heat losses from the bottom, sides, and top surfaces to the ambient air.

Closed-top vessel If the top is treated as covered, then the same overall wall coefficient U_{wall} applies to all external surfaces. The total vessel heat loss is

$$Q_{\text{loss,closed}} = U_{\text{wall}} A_{\text{total}} (T_{\text{bin}} - T_\infty), \tag{70}$$

where

$$A_{\text{total}} = A_{\text{side}} + A_{\text{bottom}} + A_{\text{top}}. \tag{71}$$

Since

$$A_{\text{side}} = 2HL + 2HW, \qquad A_{\text{bottom}} = LW, \qquad A_{\text{top}} = LW, \tag{72}$$

this may also be written as

$$Q_{\text{loss,closed}} = U_{\text{wall}} \left(2HL + 2HW + 2LW \right) (T_{\text{bin}} - T_\infty). \tag{73}$$

Open-top vessel If the top is treated as open, then the sides and bottom still lose heat through the insulated vessel wall with coefficient U_{wall}, while the exposed top loses heat directly to ambient by convection with coefficient h_{top}. The total vessel heat loss is therefore

$$Q_{\text{loss,open}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}})(T_{\text{bin}} - T_\infty) + h_{\text{top}}A_{\text{top}}(T_{\text{bin}} - T_\infty). \quad (74)$$

Substituting the areas gives

$$Q_{\text{loss,open}} = \left[U_{\text{wall}}(2HL + 2HW + LW) + h_{\text{top}}(LW)\right](T_{\text{bin}} - T_\infty). \quad (75)$$

Equivalent overall conductance It is often convenient to write these in terms of an overall conductance UA.

For a closed top,

$$(UA)_{\text{closed}} = U_{\text{wall}}A_{\text{total}}. \quad (76)$$

For an open top,

$$(UA)_{\text{open}} = U_{\text{wall}}(A_{\text{side}} + A_{\text{bottom}}) + h_{\text{top}}A_{\text{top}}. \quad (77)$$

Hence,

$$Q_{\text{loss}} = (UA)(T_{\text{bin}} - T_\infty), \quad (78)$$

with the appropriate form of UA selected according to whether the vessel top is modeled as closed or open.

0.2 Covered Top with a Small Vent Hole

In addition to the limiting cases of a fully covered top and a fully open top, the thermal model includes an intermediate case in which the bin top is covered except for a small circular vent opening that allows the injected air to escape. Let the total top area be

$$A_{\text{top}} = LW, \quad (79)$$

and let the vent consist of N_{vent} identical circular holes of diameter d_{vent}. The total open vent area is then

$$A_{\text{vent}} = N_{\text{vent}}\frac{\pi d_{\text{vent}}^2}{4}, \quad (80)$$

with the covered portion of the top given by

$$A_{\text{top,cov}} = A_{\text{top}} - A_{\text{vent}}. \quad (81)$$

The covered portion of the top is treated in the same manner as the insulated vessel wall, using the overall wall coefficient U_{wall}. The vent opening is treated as a locally open horizontal surface that exchanges heat directly with the greenhouse air by natural convection. For the vent opening, the characteristic length is taken as

$$L_{c,\text{vent}} = \frac{A_{\text{vent}}}{P_{\text{vent}}}, \quad (82)$$

where the total vent perimeter is

$$P_{\text{vent}} = N_{\text{vent}} \pi d_{\text{vent}}. \tag{83}$$

Using the same horizontal free-convection relation adopted for the open-top case, the vent-opening convection coefficient is written as

$$h_{\text{top,vent}} = \frac{Nu\, k}{L_{c,\text{vent}}}. \tag{84}$$

With this mixed covered-open boundary condition, the top-node ambient-loss conductance becomes

$$UA_{\text{top,covered+vent}} = U_{\text{wall}} \left(\frac{1}{2} A_{\text{side}} + A_{\text{top,cov}} \right) + h_{\text{top,vent}} A_{\text{vent}}. \tag{85}$$

The total vessel conductance to ambient is therefore

$$(UA)_{\text{covered+vent}} = U_{\text{wall}} \left(A_{\text{side}} + A_{\text{bottom}} + A_{\text{top,cov}} \right) + h_{\text{top,vent}} A_{\text{vent}}. \tag{86}$$

The corresponding heat loss is

$$Q_{\text{loss,covered+vent}} = (UA)_{\text{covered+vent}} \left(T_{\text{bin}} - T_{\infty} \right). \tag{87}$$

This formulation reduces to the fully covered-top model when $A_{\text{vent}} \to 0$, and it approaches the open-top model as $A_{\text{vent}} \to A_{\text{top}}$. In the present implementation, the vent is modeled as a thermal opening only; separate effects such as moisture transport, buoyancy-driven exchange through the vent, or forced vent exhaust are not added as independent loss terms.

0.3 Perforation-Driven Moisture Transfer, Water Make-Up, and Computational Selection

This section documents the moisture-transfer model, the constrained NSGA-II/Pareto heating-selection workflow, and the numerical procedure used by the heater-design code. The governing water-property relations follow the IAPWS saturation ancillary equations, the hole-scale convection coefficient is obtained from the Churchill–Bernstein crossflow correlation, and the heat–mass transfer link is written with the Lewis relation. General energy balances and continuity relations follow standard heat- and mass-transfer analysis. In the source-code commentary these formula families are tagged as IAPWS saturation relations [R9], Churchill–Bernstein crossflow [R1], the Lewis relation [R10], standard energy/continuity balances [R5], and the Frederickson et al. particle-size data [R11].

0.3.1 Implemented Nusselt Correlations

The code uses four distinct Nusselt-correlation branches, and the LaTeX documentation should state them explicitly because they are not interchangeable.

1. Crossflow over the heater wire and perforation holes For both the heated wire and the perforation-scale crossflow, the code uses the Churchill–Bernstein cylinder-in-crossflow relation [R1]:

$$Nu = 0.3 + \left(\frac{0.62\, Re^{1/2} Pr^{1/3}}{\left[1 + (0.4/Pr)^{2/3}\right]^{1/4}} \right) \left[1 + \left(\frac{Re}{282000} \right)^{5/8} \right]^{4/5}. \tag{88}$$

This is then converted to a convection coefficient by

$$h = \frac{Nu\, k}{D}, \tag{89}$$

with $D = D_{\text{wire}}$ for wire cooling and $D = d_h$ for the perforation-hole moisture calculation.

2. Internal convection in the heater tube and aeration tubes The code uses a piecewise internal-flow model. For laminar developing flow, it uses the Hausen-type entrance correction [R5]:

$$Gz = Re\, Pr\, \frac{D}{L}, \tag{90}$$

$$Nu = 3.66 + \frac{0.0668\, Gz}{1 + 0.04\, Gz^{2/3}}, \qquad Re < 2300. \tag{91}$$

For turbulent flow, it uses the Gnielinski correlation [R2]:

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)}, \qquad Re \geq 2300. \tag{92}$$

The implementation enforces

$$Nu \geq 3.66 \tag{93}$$

as a numerical lower bound. The same Nu branch is used for both the heater tube and the aeration tubes, and the same Churchill friction factor [R3] supplies the f entering the Gnielinski relation.

3. Natural convection from the vessel vertical walls For the external vertical walls, the code evaluates the Rayleigh number

$$Ra = \frac{g\beta \Delta T L^3}{\nu \alpha}, \tag{94}$$

then uses the Churchill–Chu vertical-plate correlation [R4]:

$$Nu = \left[0.825 + \frac{0.387\, Ra^{1/6}}{\left(1 + (0.492/Pr)^{9/16}\right)^{8/27}} \right]^2. \tag{95}$$

The resulting free-convection coefficient is

$$h = \frac{Nu\, k}{L}. \tag{96}$$

4. Natural convection from horizontal top or vent surfaces For the open top or vent opening, the code uses McAdams-style horizontal-plate relations [R5]. For a heated upward-facing surface (the usual cooling/heating case when the bed surface is warmer than ambient),

$$Nu = 0.54\, Ra^{1/4}, \qquad Ra \leq 10^7, \tag{97}$$

$$Nu = 0.15\, Ra^{1/3}, \qquad Ra > 10^7. \tag{98}$$

For the opposite orientation branch used when the surface is cooler than the surroundings, the code uses

$$Nu = 0.27\, Ra^{1/4}. \tag{99}$$

Again,

$$h = \frac{Nu\, k}{L}. \tag{100}$$

These four Nusselt branches are the complete set used by the present implementation. Everything else in the thermal model uses those resulting heat-transfer coefficients rather than introducing additional convection correlations.

0.3.2 Perforation Continuity and Hole Exit Velocity

Let the total imposed inlet volumetric flow rate be $\dot{V}_{\text{tot}}$, measured at the inlet state. With N_{tubes} parallel aeration tubes, the inlet flow per tube is

$$\dot{V}_{\text{tube,in}} = \frac{\dot{V}_{\text{tot}}}{N_{\text{tubes}}}. \tag{101}$$

Using the inlet air density ρ_{in}, the inlet mass flow rate per tube is

$$\dot{m}_{\text{tube,in}} = \rho_{\text{in}} \dot{V}_{\text{tube,in}}. \tag{102}$$

The model assumes that a fraction ϕ_{rel} of the tube flow is released into the bed through perforations, so that

$$\dot{m}_{\text{rel}} = \phi_{\text{rel}} \dot{m}_{\text{tube,in}}. \tag{103}$$

If each tube contains N_h holes of diameter d_h, the area per hole is

$$A_h = \frac{\pi d_h^2}{4}, \tag{104}$$

and the total open area per tube is

$$A_{h,\text{tot}} = N_h A_h. \tag{105}$$

Because the released air is evaluated at the local heated-air state, its volumetric release rate is

$$\dot{V}_{\text{rel}} = \frac{\dot{m}_{\text{rel}}}{\rho_{\text{air}}(T_{\text{air}}, p)}. \tag{106}$$

The mean released flow per hole is therefore

$$\dot{V}_{\text{hole}} = \frac{\dot{V}_{\text{rel}}}{N_h}, \tag{107}$$

and the mean hole exit velocity is

$$u_h = \frac{\dot{V}_{\text{rel}}}{N_h A_h}. \tag{108}$$

Why u_h varies with voltage. At fixed imposed inlet flow $\dot{V}_{\text{tot}}$, the inlet mass flow rate is nearly fixed by the inlet density. However, increasing the branch voltage increases electrical power, which raises the heater outlet temperature T_{air}. Since the released-air density $\rho_{\text{air}}(T_{\text{air}}, p)$ decreases as the air is heated, the same released mass flow occupies a larger volume. Consequently $\dot{V}_{\text{rel}}$ and hence u_h increase with voltage even when the imposed inlet flow axis is held constant.

0.3.3 Particle-Size-Based Wetted-Area Estimate

The current implementation no longer uses the bed plan area as the default evaporation area. Instead, it derives an external particle-surface area from the particle-size distribution reported in Table 3 of Frederickson et al. [R11]. The reported matured-material size fractions are:

$$\text{windrow: } 9.8\% \ (> 40 \text{ mm}),\ 26.6\% \ (20\text{–}40 \text{ mm}),\ 26.6\% \ (10\text{–}20 \text{ mm}),\ 37.0\% \ (< 10 \text{ mm}), \tag{109}$$

$$\text{vermicompost: } 7.9\% \ (> 40 \text{ mm}),\ 6.9\% \ (20\text{–}40 \text{ mm}),\ 19.8\% \ (10\text{–}20 \text{ mm}),\ 65.3\% \ (< 10 \text{ mm}). \tag{110}$$

The code forms the blended fractions

$$w_i = \frac{1}{2}\left(w_i^{\text{windrow}} + w_i^{\text{vermi}}\right), \tag{111}$$

then assigns each bin an equivalent representative particle diameter

$$d_i \in \{60,\ 30,\ 15,\ 5\} \text{ mm} \tag{112}$$

for the classes > 40, 20–40, 10–20, and < 10 mm respectively. The closed bins use interval midpoints, while the open-ended > 40 mm bin is represented by 60 mm, i.e. one adjacent-bin width above the lower cut.

The area calculation then uses the Sauter mean diameter for equal-density equivalent spheres:

$$\frac{1}{d_{32}} = \sum_i \frac{w_i}{d_i}, \tag{113}$$

so that the external surface area per occupied bed volume is

$$a_{\text{ext}} = \frac{6}{d_{32}}. \tag{114}$$

This follows directly from the sphere area-to-volume ratio $A/V = 6/d$ and is the simplest particle-size-based estimate that still responds correctly to the observed shift toward finer material.

Let the filled bed volume be

$$V_{\text{fill}} = LWHf_{\text{fill}}. \tag{115}$$

The effective wetted area is then

$$A_{\text{wet}} = f_{\text{access}}\, a_{\text{ext}}\, V_{\text{fill}} = f_{\text{access}}\, \frac{6V_{\text{fill}}}{d_{32}}, \tag{116}$$

where $f_{\text{access}} \in [0, 1]$ is the accessible-surface fraction used to represent the share of the external particle area that is simultaneously wet and exposed to the flowing air.

For the current default assumptions,

$$w_i = \{0.0885,\ 0.1675,\ 0.2320,\ 0.5115\}, \tag{117}$$

which gives

$$d_{32} \approx 8.01 \text{ mm}, \qquad a_{\text{ext}} \approx 749 \text{ m}^2/\text{m}^3. \tag{118}$$

With

$$V_{\text{fill}} = 1.22 \times 0.61 \times 0.61 \times 0.80 \approx 0.363 \text{ m}^3, \tag{119}$$

and $f_{\text{access}} = 1$, the default effective wetted area becomes

$$A_{\text{wet}} \approx 272 \text{ m}^2. \tag{120}$$

This is intentionally an *external agglomerate surface* estimate, not a true microscopic pore-wall area. The equal-density assumption means the reported mass fractions are treated as proxies for volume fractions. That is appropriate for a simple design-stage model because it preserves the dominant effect of particle refinement on interfacial area without claiming a full microstructural reconstruction of the compost matrix.

The code distributes this area uniformly across the aeration segments, so the effective wetted area attached to a single segment is

$$A_{\text{seg}} = \frac{A_{\text{wet}}}{N_{\text{tubes}} N_{\text{aer}}}. \tag{121}$$

0.3.4 Water Saturation Properties and Latent Heat

The vapor-density driving force is built from saturation properties of water. First, the saturation pressure is computed from the IAPWS ancillary form

$$\ln\left(\frac{p_{\text{sat}}}{p_c}\right) = \frac{T_c}{T} \sum_{i=1}^{6} a_i \tau^{b_i}, \qquad \tau = 1 - \frac{T}{T_c}, \tag{122}$$

where T_c and p_c are the critical temperature and pressure of water.

The saturated liquid density is written as

$$\rho_{\ell,\mathrm{sat}} = \rho_c \left(1 + \sum_{i=1}^{6} \beta_i \tau^{n_i}\right), \tag{123}$$

and the saturated vapor density is

$$\rho_{v,\mathrm{sat}} = \rho_c \exp\left(\sum_{i=1}^{6} \gamma_i \tau^{m_i}\right), \tag{124}$$

with ρ_c the critical density of water.

The saturation vapor density used in the mass-transfer driving force is then

$$\rho_{v,\mathrm{sat}}(T) = \frac{p_{\mathrm{sat}}(T)}{R_v T}, \tag{125}$$

where R_v is the specific gas constant of water vapor.

The latent heat of vaporization is obtained from the Clausius–Clapeyron relation along the saturation line:

$$h_{fg}(T) = T \frac{dp_{\mathrm{sat}}}{dT} \left(\frac{1}{\rho_{v,\mathrm{sat}}(T)} - \frac{1}{\rho_{\ell,\mathrm{sat}}(T)}\right). \tag{126}$$

0.3.5 Hole-Scale Convection and Mass-Transfer Coefficients

The model treats each perforation as a characteristic crossflow scale for heat and mass transfer. Using the local heated-air properties, the hole-scale Reynolds number is

$$\mathrm{Re}_h = \frac{\rho_{\mathrm{air}} u_h d_h}{\mu_{\mathrm{air}}}, \tag{127}$$

and the Nusselt number is evaluated with the Churchill–Bernstein correlation:

$$\mathrm{Nu}_h = 0.3 + \frac{0.62 \mathrm{Re}_h^{1/2} \mathrm{Pr}^{1/3}}{\left[1 + (0.4/\mathrm{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\mathrm{Re}_h}{282000}\right)^{5/8}\right]^{4/5}. \tag{128}$$

The corresponding convective coefficient is

$$h_{\mathrm{evap}} = \mathrm{Nu}_h \frac{k_{\mathrm{air}}}{d_h}. \tag{129}$$

The heat–mass transfer analogy is then used in the form

$$h_m = \frac{h_{\mathrm{evap}}}{\rho_{\mathrm{air}} c_{p,\mathrm{air}} \mathrm{Le}^{2/3}}, \tag{130}$$

where Le is the Lewis factor. In the present implementation, Le is taken near unity for the air–water system.

0.3.6 Evaporation Driving Force and Three Limiting Rates

Let the prescribed effective wet-surface temperature be T_s and the inlet relative humidity be ϕ_∞. The model constructs the vapor-density driving force from

$$\rho_{v,s} = \rho_{v,\mathrm{sat}}(T_s), \tag{131}$$

$$\rho_{v,\infty} = \phi_\infty \rho_{v,\mathrm{sat}}(T_\infty), \tag{132}$$

so that

$$\Delta\rho_v = \max(\rho_{v,s} - \rho_{v,\infty}, 0)\,. \tag{133}$$

For each aeration-tube segment, the model computes three candidate evaporation rates and accepts the smallest of the three.

Diffusion-limited rate. If A_{seg} is the effective wetted area assigned to a single segment, then

$$\dot{m}_{\mathrm{diff}} = h_m A_{\mathrm{seg}} \Delta\rho_v. \tag{134}$$

Carrying-capacity limit of the released air parcel. If $\dot{V}_{\mathrm{rel,seg}}$ is the released volumetric flow rate for that segment, then the maximum vapor uptake permitted by the released air volume is

$$\dot{m}_{\mathrm{cap}} = \dot{V}_{\mathrm{rel,seg}} \Delta\rho_v. \tag{135}$$

Heat-limited evaporation rate. The released jet also has a finite sensible heat content. The model defines the available sensible power for evaporation as

$$q_{\mathrm{avail}} = \max(\dot{m}_{\mathrm{rel,seg}} c_{p,\mathrm{air}} (T_{\mathrm{air}} - T_{\mathrm{bed}}), 0)\,, \tag{136}$$

which leads to

$$\dot{m}_{\mathrm{heat}} = \frac{q_{\mathrm{avail}}}{h_{fg}}. \tag{137}$$

The accepted segment evaporation rate is

$$\dot{m}_{\mathrm{evap,seg}} = \min(\dot{m}_{\mathrm{diff}}, \dot{m}_{\mathrm{cap}}, \dot{m}_{\mathrm{heat}})\,, \tag{138}$$

and the corresponding latent heat sink is

$$q_{\mathrm{evap,seg}} = \dot{m}_{\mathrm{evap,seg}} h_{fg}. \tag{139}$$

Unit consistency. The three candidate evaporation rates are dimensionally consistent:

- $h_m A \Delta\rho_v$ has units $(\mathrm{m/s})(\mathrm{m}^2)(\mathrm{kg/m}^3) = \mathrm{kg/s}$;
- $\dot{V}_{\mathrm{rel,seg}} \Delta\rho_v$ has units $(\mathrm{m}^3/\mathrm{s})(\mathrm{kg/m}^3) = \mathrm{kg/s}$; and
- $q_{\mathrm{avail}}/h_{fg}$ has units $(\mathrm{J/s})/(\mathrm{J/kg}) = \mathrm{kg/s}$.

Therefore $\dot{m}_{\mathrm{diff}}$, $\dot{m}_{\mathrm{cap}}$, $\dot{m}_{\mathrm{heat}}$, and $\dot{m}_{\mathrm{evap,seg}}$ are all compared in consistent kg/s units.

Why water loss and latent load vary with voltage. Voltage increases electrical heating, which typically raises T_{air} at fixed inlet flow. This has three effects:

1. It lowers the local air density, thereby increasing $\dot{V}_{\text{rel}}$ and u_h.
2. It increases Re_h and hence usually increases h_{evap} and h_m.
3. It increases the sensible heating term q_{avail}, which raises the heat-limited evaporation cap.

Accordingly, the water-loss and latent-load maps generally increase with voltage, although the final contours are shaped by the minimum operator among the diffusion, carrying-capacity, and heat-limited branches.

0.3.7 Radial Air-Cooling Profile and Worm-Safe Distance

The standalone Python study now adds a radial post-processing model to estimate how rapidly the released warm air cools as it moves outward from the aeration tube into the wet bed. This is not a new CFD solver; it is a finite top-hat plume model built from the same moisture and wetted-area quantities already computed by the main design code.

Let x denote the radial distance measured outward from the aeration-tube outer surface, and let

$$r_0 = \frac{D_{\text{tube,OD}}}{2} \tag{140}$$

be the tube outer radius. The released mass flow rate per unit tube length is

$$\dot{m}'_{\text{rel}} = \frac{\phi_{\text{rel}} \dot{m}_{\text{tube,in}}}{L_{\text{aer}}}. \tag{141}$$

Using the previously derived wetted area, the corresponding interfacial area density is

$$a_{\text{wet}} = \frac{A_{\text{wet}}}{V_{\text{fill}}}. \tag{142}$$

Here T_{sink} is the clamped wet-surface temperature used as the local cooling asymptote:

$$T_{\text{sink}} = \min(\max(T_{s,\text{set}}, T_{\text{bed,ref}}), T_{\text{out}}) \,. \tag{143}$$

Instead of assuming that the full local wetted area is active immediately, the plume model assumes the warm released air occupies a finite plume of radius $b(x)$ that grows with distance:

$$b(x) = \min(b_{\max},\, b_0 + \alpha x) \,. \tag{144}$$

Here b_0 is the initial plume radius, usually taken as the hole radius, α is the plume spreading rate, and $b_{\max}$ is an optional cap. These are exposed in the JSON file through the `radial_profile` block.

Unlike the earlier approximation, the present implementation does *not* collapse latent cooling into a single fixed equivalent coefficient. Instead it re-solves the plume temperature and vapor state along the radial path. The contacted wetted area per unit radial distance is

$$A'_{\mathrm{wet}}(x) = \pi a_{\mathrm{wet}} b(x)^2. \tag{145}$$

The sensible cooling term is

$$q'_{\mathrm{sens}} = h_{\mathrm{sens}} A'_{\mathrm{wet}}(x) \left(T - T_{\mathrm{sink}}\right), \tag{146}$$

and the temperature equation is

$$\dot{m}'_{\mathrm{rel}} c_{p,\mathrm{air}} \frac{dT}{dx} = -\left(q'_{\mathrm{sens}} + q'_{\mathrm{lat}}\right). \tag{147}$$

The vapor-density state is updated from a Lewis-relation mass-transfer model written with local air properties:

$$h_m(x) = \frac{h_{\mathrm{evap}}}{\rho_{\mathrm{air}}(T)\, c_{p,\mathrm{air}}\, \mathrm{Le}^{2/3}}. \tag{148}$$

Using the same wet-surface temperature clamp as the main moisture model, the local diffusion-driven evaporation rate per unit radial distance is

$$\dot{m}'_{\mathrm{evap,pot}} = h_m(x)\, A'_{\mathrm{wet}}(x)\, \left(\rho_{v,\mathrm{sat}}(T_s) - \rho_{v,\mathrm{bulk}}(x)\right). \tag{149}$$

The plume humidity is then advanced explicitly so that the bulk vapor density cannot exceed local saturation at the updated plume temperature. In this way, the latent sink weakens automatically as the plume both cools and humidifies.

The corresponding latent heat term is

$$q'_{\mathrm{lat}} = \dot{m}'_{\mathrm{evap}} h_{fg}(T_s), \tag{150}$$

and the total temperature drop is limited so that the plume cannot cool below T_{sink} within a single marching step.

This is still a reduced-order model rather than CFD, but it is more physically consistent than the earlier constant-h_{eff} approximation because the radial evaporation rate now responds directly to the local plume temperature and local vapor build-up.

The solution is obtained numerically on the radial grid used for the plot; the earlier closed-form expression for $T(x)$ is no longer used because the latent term is now state-dependent.

If the worm-safety threshold is $T_{\mathrm{safe}} = 27°\mathrm{C}$, the safe offset from the tube outer surface is the first x satisfying

$$T(x) = T_{\mathrm{safe}}. \tag{151}$$

In the code this crossing is found by interpolation on the numerically evaluated profile.

This construction is intentionally modest in scope. It still does *not* resolve individual jets or local tortuous pore paths, but it is more physically consistent than the earlier immediate-full-area sink because it delays plume coupling to the full local wetted surface until the warm-air core has spread through a larger pore volume.

0.3.8 Tube Layout, Safe-Distance Exclusion Cylinders, and Habitable Volume

The standalone study also converts the worm-safe radial cutoff into a geometric estimate of the bed volume that is effectively too warm for worm occupation. This calculation is performed in the bed cross section and then extruded over the aeration-tube length.

Let the filled-bed height be

$$H_{\text{fill}} = f_{\text{fill}} H, \tag{152}$$

so that the filled-bed cross-sectional domain is

$$\Omega_{\text{fill}} = \{(x, y) : 0 \leq x \leq W,\ 0 \leq y \leq H_{\text{fill}}\}\,. \tag{153}$$

The corresponding filled-bed volume is

$$V_{\text{fill}} = LWH_{\text{fill}}. \tag{154}$$

If the user does not specify tube-center locations explicitly, the current implementation uses the requested rule-based layout for $1 \leq N_{\text{tubes}} \leq 8$ inside the *filled-bed* rectangle. Let

$$x_L = f_x W, \qquad x_R = (1 - f_x)W, \tag{155}$$

$$y_B = f_y H_{\text{fill}}, \qquad y_T = (1 - f_y)H_{\text{fill}}, \tag{156}$$

with default offsets $f_x = f_y = 0.10$. The rules are:

- 1 tube: $(W/2,\ y_B)$;
- 2 tubes: bottom-left and bottom-right;
- 3 tubes: three equally spaced across the bottom row from x_L to x_R;
- 4 tubes: the four corners (x_L, y_B), (x_R, y_B), (x_L, y_T), (x_R, y_T);
- 5 tubes: the 3-bottom-row pattern plus the two top corners;
- 6 tubes: three equally spaced vertically on the left side and three on the right side;
- 7 tubes: the four corners plus left-mid, right-mid, and bottom-center; and
- 8 tubes: the 7-tube pattern plus a central tube at $(W/2,\ H_{\text{fill}}/2)$.

If explicit positions are supplied in the JSON file, those values replace the rule-based pattern directly.

The center-to-center pitch between neighboring tubes is

$$p_i = x_{i+1} - x_i, \tag{157}$$

and the corresponding clear gap between the outer tube surfaces is

$$g_i = p_i - D_{\text{tube,OD}}. \tag{158}$$

Using the radial safe offset x_{safe} from the previous section, the safe-distance exclusion radius around each tube is

$$R_{\text{exc}} = r_0 + x_{\text{safe}}. \tag{159}$$

Thus each actual pipe occupies a circular cross section of radius r_0, while the worm-exclusion region occupies a larger circle of radius R_{exc} centered on the same tube axis.

To account correctly for overlap between neighboring exclusion envelopes, the study does *not* sum four independent cylinders. Instead, it evaluates the union area in cross section. Define the pipe-indicator function

$$I_{\text{pipe}}(x, y) = \begin{cases} 1, & \exists i \text{ such that } (x - x_i)^2 + (y - y_i)^2 \leq r_0^2, \\ 0, & \text{otherwise,} \end{cases} \tag{160}$$

and the exclusion-indicator function

$$I_{\text{exc}}(x, y) = \begin{cases} 1, & \exists i \text{ such that } (x - x_i)^2 + (y - y_i)^2 \leq R_{\text{exc}}^2, \\ 0, & \text{otherwise.} \end{cases} \tag{161}$$

The corresponding overlap-aware union areas are

$$A_{\text{pipe,union}} = \iint_{\Omega_{\text{fill}}} I_{\text{pipe}}(x, y)\, dA, \tag{162}$$

and

$$A_{\text{exc,union}} = \iint_{\Omega_{\text{fill}}} I_{\text{exc}}(x, y)\, dA. \tag{163}$$

Since the aeration tubes run parallel to the vessel length, the associated volumes are obtained by extrusion:

$$V_{\text{pipe}} = A_{\text{pipe,union}} L_{\text{aer}}, \tag{164}$$

$$V_{\text{exc}} = A_{\text{exc,union}} L_{\text{aer}}. \tag{165}$$

The pipe-solid volume is therefore V_{pipe}, the worm-unsafe shell volume outside the pipe metal but inside the safe-distance envelope is

$$V_{\text{unsafe,shell}} = V_{\text{exc}} - V_{\text{pipe}}, \tag{166}$$

and the remaining habitable bed volume is

$$V_{\text{hab}} = V_{\text{fill}} - V_{\text{exc}}. \tag{167}$$

The current implementation also adds a simple wall-return correction for the left, right, and bottom vessel sheets. If the plume reaches a wall, the wall contact temperature is sampled from the radial plume solution at the wall-offset distance. Using the wall stack already present in the main heat loss model, the inside wall temperature is estimated from the 1D resistance network

$$q''_{\text{wall}} = \frac{T_{\text{plume,wall}} - T_\infty}{\frac{1}{h_{\text{in}}} + \frac{t_{\text{sheet}}}{k_{\text{sheet}}} + \frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_{\text{ext}}}}, \tag{168}$$

$$T_{\text{wall,in}} = T_{\text{plume,wall}} - \frac{q''_{\text{wall}}}{h_{\text{in}}}. \tag{169}$$

Assuming the steel sheet redistributes this temperature approximately uniformly along the corresponding wall, a 1D linear wall-normal soil profile is then used to estimate an additional unsafe band depth. Those wall bands are merged with the exclusion cylinders through the same union-area calculation, so the final unsafe volume reported by the code is the union of pipe cylinders, safe-distance cylinders, and wall-return bands:

$$V_{\text{unsafe,total}} = V_{\text{union}}. \tag{170}$$

The code also reports explicit overlap corrections by comparing the union areas above with the naive no-overlap sums:

$$A_{\text{pipe,sum}} = N_{\text{tubes}} \pi r_0^2, \qquad A_{\text{exc,sum}} = N_{\text{tubes}} \pi R_{\text{exc}}^2. \tag{171}$$

The removed double-counted areas are

$$A_{\text{pipe,ov}} = A_{\text{pipe,sum}} - A_{\text{pipe,union}}, \tag{172}$$

$$A_{\text{exc,ov}} = A_{\text{exc,sum}} - A_{\text{exc,union}}, \tag{173}$$

with volume counterparts

$$V_{\text{pipe,ov}} = A_{\text{pipe,ov}} L_{\text{aer}}, \qquad V_{\text{exc,ov}} = A_{\text{exc,ov}} L_{\text{aer}}. \tag{174}$$

These overlap terms are useful because they show how much the naive "sum of independent cylinders" estimate would over-predict the inaccessible or unsafe volume when neighboring exclusion envelopes intersect.

Numerically, the present implementation evaluates $A_{\text{pipe,union}}$ and $A_{\text{exc,union}}$ on a midpoint grid over Ω_{fill}. If N_g points are used in each direction, the area estimator is

$$A \approx \left(\frac{W H_{\text{fill}}}{N_g^2} \right) \sum_{j=1}^{N_g} \sum_{k=1}^{N_g} I(x_j^*, y_k^*), \tag{175}$$

where (x_j^*, y_k^*) are the cell midpoints. The resulting union area is then multiplied by L_{aer} to obtain the reported volume. This is sufficient here because the cross section contains only a small number of circles and the output is intended as an engineering habitat-volume estimate rather than an exact computational-geometry benchmark.

0.3.9 Aeration-Tube Energy Balance with Moisture Sink

For each aeration-tube segment, the wall-mediated sensible transfer is

$$q_{\text{wall,seg}} = U_{\text{aer}} P_{\text{out}} \Delta x (T_{\text{air}} - T_{\text{bed}}), \tag{176}$$

and the direct sensible jet contribution is

$$q_{\text{jet,seg}} = \dot{m}_{\text{rel,seg}} c_{p,\text{air}} (T_{\text{air}} - T_{\text{bed}}). \tag{177}$$

With evaporation included explicitly, the useful heat delivered to the bed by one aeration branch is

$$Q_{\text{bed,branch}} = \sum_{\text{seg}} \left(q_{\text{wall,seg}} + q_{\text{jet,seg}} - q_{\text{evap,seg}} \right). \tag{178}$$

For N_{tubes} identical parallel tubes,

$$Q_{\text{bed,total}} = N_{\text{tubes}} Q_{\text{bed,branch}}. \tag{179}$$

The total modeled water make-up requirement is

$$\dot{m}_{\text{water,day}} = 86400\, N_{\text{tubes}} \sum_{\text{seg}} \dot{m}_{\text{evap,seg}}, \tag{180}$$

reported in kg/24 h. This quantity is treated as the sprinkler replacement requirement and is also used directly as a design constraint.

0.4 Constrained NSGA-II Pareto Selection and Design-Point Reporting

The MATLAB model still generates the physics by evaluating a discrete sweep of pitch, voltage, flow, and geometry candidates. The current standalone Python report layer no longer chooses the final reported heating design point by direct lexicographic ranking of the full sweep. Instead, it applies constrained NSGA-II to the already evaluated discrete candidate set and then reports one point from the feasible Pareto set.

0.4.1 Candidate Objectives and Constraint Residuals

For each operating point, the code computes the full-power two-node bed temperatures T_b and T_t, their spread

$$\Delta T_{\text{bed}} = |T_b - T_t|, \tag{181}$$

the colder-node temperature

$$T_{\text{cold}} = \min(T_b, T_t), \tag{182}$$

and the duty ratio required to meet the lumped design heat loss,

$$\text{dutyNeeded} = \frac{Q_{\text{req}}}{Q_{\text{bed,total}}}. \tag{183}$$

The current heating-objective vector is

$$\mathbf{F} = [P_{\text{tot}}, -T_{\text{cold}}, \Delta T_{\text{bed}}], \tag{184}$$

so the reported Pareto front minimizes total electrical power, maximizes the colder-node bed temperature, and minimizes bottom–top spread.

The inequality-constraint residual vector is

$$\mathbf{G} = [G_1, G_2, \ldots, G_{11}], \tag{185}$$

with

$$G_1 = \max(I_{\text{tot}} - I_{\max}, 0), \tag{186}$$
$$G_2 = \max(T_{\text{wire}} - T_{\text{wire,max}}, 0), \tag{187}$$
$$G_3 = \max(T_{\text{air,out}} - T_{\text{air,max}}, 0), \tag{188}$$
$$G_4 = \max(\Delta P - \Delta P_{\max}, 0), \tag{189}$$
$$G_5 = \max(\dot{m}_{\text{water,day}} - \dot{m}_{\text{water,max}}, 0), \tag{190}$$
$$G_6 = \max(u_{\text{hole}} - u_{\text{hole,max}}, 0), \tag{191}$$
$$G_7 = \max(T_{b,\min} - T_b, 0), \tag{192}$$
$$G_8 = \max(T_{t,\min} - T_t, 0), \tag{193}$$
$$G_9 = \max(\Delta T_{\text{bed}} - \Delta T_{\max}, 0), \tag{194}$$
$$G_{10} = \max(\text{dutyNeeded} - 1, 0), \tag{195}$$
$$G_{11} = \max(-Q_{\text{bed,total}}, 0). \tag{196}$$

Thus a feasible heating point is one for which every component of $\mathbf{G}$ is zero. In particular, the reported design point must satisfy the current, wire-temperature, outlet-air, pressure-drop, water-loss, hole-velocity, bottom-temperature, top-temperature, spread, and full-load heating requirements simultaneously.

0.4.2 NSGA-II on the Discrete Candidate Grid

The current implementation uses NSGA-II through `pymoo`, but it does not replace the underlying physics solve. Instead, the algorithm operates on integer-valued indices into the already evaluated discrete pitch, voltage, and flow arrays. The present default population size in the JSON configuration is

$$N_{\text{pop}} = 50. \tag{197}$$

This choice preserves the rigor of the full MATLAB candidate evaluation while replacing the old single-score final-selection step with a proper multiobjective feasible-Pareto search.

0.4.3 Reported Design Point from the Feasible Pareto Set

NSGA-II returns a feasible Pareto candidate set in the objective space above. The standalone summary and report then choose one displayed design point from that feasible Pareto set by a user-configurable report-priority order. In the current default configuration, the reported-point ordering is:

1. maximize the colder-node temperature T_{cold};
2. minimize total electrical power P_{tot};
3. minimize bottom–top spread ΔT_{bed};
4. use mean bed temperature only as a later tie-breaker; and
5. use residual heat shortfall only as a final tie-breaker.

Accordingly, the reported design point is *not* an arbitrary weighted sum. It is the first point in the feasible Pareto set after applying the configured report-priority order.

0.4.4 Fallback Reference Point

If no fully feasible Pareto point exists, the code reports no recommended heating design point. In that case, it retains a best-available reference point only for diagnostic plots and fallback summaries. That fallback point is still based on a lexicographic least-violation ranking of the full sweep, so the user can see which constraint is blocking feasibility without falsely presenting an infeasible point as a final recommendation.

0.4.5 Where the Pareto Workflow Is Used

The constrained-NSGA-II heating workflow now drives:

1. the reported recommended point for the active vessel configuration;
2. the configuration-specific recommended point shown in the vessel comparison table; and
3. the active-configuration Pareto-candidate table and Pareto-front plot.

The report-only duty-reference value `model_overrides.sweep.targetDuty` is still retained, but it remains a display aid for the minimum-flow duty line rather than a replacement for the full-feasibility condition above.

0.5 Numerical Procedure

0.5.1 Axial Discretization of the Heater and Aeration Tubes

The heater tube is discretized into N_{heater} axial segments, with

$$\Delta x_{\text{heater}} = \frac{L_{\text{heater}}}{N_{\text{heater}}}, \tag{198}$$

and the aeration tube is discretized into N_{aer} axial segments, with

$$\Delta x_{\text{aer}} = \frac{L_{\text{aer}}}{N_{\text{aer}}}. \tag{199}$$

In the current implementation,

$$N_{\text{heater}} = 50, \qquad N_{\text{aer}} = 40. \tag{200}$$

The solution marches explicitly from inlet to outlet through both domains.

0.5.2 Fixed-Point Iteration for Temperature-Dependent Wire Resistance

For a specified branch voltage V and total flow $\dot{V}_{\text{tot}}$, the code solves the operating point by iterating on the average wire temperature. Starting from an initial guess $\overline{T}^{(0)}_{\text{wire}}$, it evaluates the temperature-dependent wire resistance

$$\rho_{\text{wire}}(T) = \rho_{\text{wire},20}\left[1 + \alpha_{\text{wire}}(T - 20^\circ\text{C})\right], \tag{201}$$

$$R_{\text{branch}} = \rho_{\text{wire}}(T)\frac{L_{\text{wire}}}{A_{\text{wire}}}, \tag{202}$$

then computes

$$I_{\text{branch}} = \frac{V}{R_{\text{branch}}}, \qquad P_{\text{branch}} = V I_{\text{branch}}. \tag{203}$$

The heater-tube solution then returns a new mean wire temperature $\overline{T}_{\text{wire,new}}$. The code updates the iterate by damping:

$$\overline{T}^{(k+1)}_{\text{wire}} = 0.65\,\overline{T}^{(k)}_{\text{wire}} + 0.35\,\overline{T}^{(k)}_{\text{wire,new}}. \tag{204}$$

Convergence is accepted when

$$\left|\overline{T}^{(k)}_{\text{wire,new}} - \overline{T}^{(k)}_{\text{wire}}\right| < 0.05^\circ\text{C}, \tag{205}$$

or when the maximum iteration count is reached. The present implementation uses at most eight iterations per operating point.

The coefficients 0.65 and 0.35 are numerical under-relaxation factors, not physical material parameters. They are used because the coupled resistance–current–wire-temperature loop can oscillate if the full updated temperature is accepted at every iteration. A hotter guessed wire raises the resistance, which lowers current and power, which then lowers the predicted wire temperature on the next pass. Under-relaxation damps this feedback by blending the old iterate with the newly computed iterate. The chosen values are empirical but moderate: they preserve convergence speed while suppressing oscillatory overshoot over the voltage-flow sweep.

0.5.3 Heater-Tube Marching Scheme

Within each heater segment, the local generated electrical heat is

$$q_{\text{gen,seg}} = P_{\text{branch}} \frac{\Delta x_{\text{heater}}}{L_{\text{heater}}}, \tag{206}$$

the heater-shell loss to ambient is

$$q_{\text{loss,seg}} = U_{\text{heater}} P_{\text{out}} \Delta x_{\text{heater}} (T_{\text{air}} - T_{\infty}), \tag{207}$$

and the net heat retained by the airflow is

$$q_{\text{to air,seg}} = q_{\text{gen,seg}} - q_{\text{loss,seg}}. \tag{208}$$

The air temperature is advanced explicitly by

$$T_{\text{air},i+1} = T_{\text{air},i} + \frac{q_{\text{to air,seg}}}{\dot{m}_{\text{heater}} c_{p,\text{air}}}. \tag{209}$$

The local wire temperature is then reconstructed from Newton's law:

$$T_{\text{wire},i} = T_{\text{air},i} + \frac{q_{\text{gen,seg}}}{h_{\text{wire}} A_{\text{wire,seg}}}. \tag{210}$$

Here T_{∞} denotes the surrounding greenhouse ambient used by the heater-shell loss model. In the present study this is the design greenhouse air temperature (for example $-15°\text{C}$), not a separately solved local external air temperature field around the heater tube.

0.5.4 Aeration-Tube Marching Scheme with Progressive Release

The aeration tube receives the heater outlet state as its inlet. The total released mass is distributed uniformly over the aeration segments:

$$\dot{m}_{\text{rel,seg}} = \frac{\phi_{\text{rel}} \dot{m}_{\text{tube,in}}}{N_{\text{aer}}}. \tag{211}$$

At each segment, the code computes

$$q_{\text{wall,seg}} = U_{\text{aer}} P_{\text{out}} \Delta x_{\text{aer}} (T_{\text{air}} - T_{\text{bed}}), \tag{212}$$

$$q_{\text{jet,seg}} = \dot{m}_{\text{rel,seg}} c_{p,\text{air}} (T_{\text{air}} - T_{\text{bed}}), \tag{213}$$

$$q_{\text{evap,seg}} = \dot{m}_{\text{evap,seg}} h_{fg}, \tag{214}$$

and accumulates

$$Q_{\text{bed,branch}} \leftarrow Q_{\text{bed,branch}} + q_{\text{wall,seg}} + q_{\text{jet,seg}} - q_{\text{evap,seg}}. \tag{215}$$

The remaining in-tube mass flow is reduced after each segment:

$$\dot{m}_{\text{rem},i+1} = \dot{m}_{\text{rem},i} - \dot{m}_{\text{rel,seg}}, \tag{216}$$

and the bulk-tube air temperature is advanced using the wall-transfer term:

$$T_{\text{air},i+1} = T_{\text{air},i} - \frac{q_{\text{wall,seg}}}{\dot{m}_{\text{rem},i+1} c_{p,\text{air}}}. \tag{217}$$

0.5.5 Pressure-Drop Evaluation

Both the heater and aeration tubes accumulate Darcy–Weisbach friction and distributed minor losses segment by segment:

$$\Delta P_{\text{seg}} = \left(f \frac{\Delta x}{D} + \frac{K_{\text{minor}}}{N} \right) \frac{\rho u^2}{2}. \tag{218}$$

The heater-to-aeration abrupt expansion is modeled by

$$\Delta P_{\text{exp}} = \frac{\rho}{2}(u_1 - u_2)^2, \tag{219}$$

which is the standard Borda–Carnot sudden-expansion relation.

In the standalone Python heating reference-point solve, the helical wire inside the heater tube also contributes a separate reduced-order obstruction term. Rather than folding that effect into an arbitrary extra minor-loss constant, the code now evaluates a projected-area drag surrogate based on the local velocity component normal to the helix. This is an intentional reduced-order mapping of the handbook sections on tubes with wire ribbing and spiral-ribbed air heaters: those references indicate that the ribbing contributes finite aerodynamic resistance in addition to the smooth-tube Darcy wall-friction term. The active JSON configuration now exposes this closure through `coilPressureModel`, `coilDragCoefficient`, and `coilLossMultiplier` under `model_overrides.heaterTube`.

For the common distribution hardware upstream of each branch, the current implementation distinguishes five contributions:

1. an optional upstream header loss,
2. an optional header-to-splitter sudden contraction,
3. a splitter-body minor loss,
4. friction in the short branch connector, and
5. a connector-to-branch sudden contraction or sudden expansion.

If a separate large upstream header is used, the header term is

$$\Delta P_{\text{header}} = K_{\text{header}} \frac{\rho u_{\text{header}}^2}{2}. \tag{220}$$

If `headerEnabled = false`, this term is omitted.

Let the nominal splitter/connector diameter be D_c, let the branch downstream diameter be D_b, and let the short connector length be L_c. The connector-friction term is

$$\Delta P_{\text{connector}} = f_c \frac{L_c}{D_c} \frac{\rho u_c^2}{2}, \tag{221}$$

with u_c and f_c computed from the connector hydraulic diameter and the branch mass flow rate.

For a conical converging bellmouth without end wall, as tabulated by Idel'chik in Section III, Diagram 3-5, the code models a contraction from an upstream diameter D_1 to a downstream diameter $D_2 < D_1$ by defining

$$\epsilon = \frac{A_2}{A_1}, \tag{222}$$

reading the diagram coefficient $C(\epsilon, \alpha)$ at the configured cone angle α, and then using Idel'chik's equation (3-3) on the smaller-pipe velocity basis:

$$\Delta P_{\text{contr}} = K_{\text{contr}} \frac{\rho u_2^2}{2}, \tag{223}$$

with

$$\alpha = 60^\circ \quad \text{(current default)}, \qquad K_{\text{contr}} = C(\epsilon, \alpha) \left(\frac{1}{\epsilon} - 1 \right). \tag{224}$$

The current implementation interpolates the tabulated C values from Diagram 3-5 over

$$\epsilon \in \{0.025, 0.050, 0.075, 0.10, 0.15, 0.25, 0.60, 1.0\} \tag{225}$$

and

$$\alpha \in \{0^\circ, 10^\circ, 20^\circ, 30^\circ, 40^\circ, 60^\circ, 100^\circ, 140^\circ, 180^\circ\}. \tag{226}$$

This form is used for the optional header-to-splitter contraction and, when applicable, for the connector-to-downstream contraction.

For a sudden expansion from D_1 to $D_2 > D_1$, the code uses the Idel'chik Section IV sudden-expansion coefficient, written on the upstream-velocity basis as

$$\Delta P_{\text{exp}} = K_{\text{exp}} \frac{\rho u_1^2}{2}, \qquad K_{\text{exp}} = \left(1 - \frac{A_1}{A_2} \right)^2, \tag{227}$$

which is algebraically equivalent to the Borda–Carnot form already written above and is the same coefficient quoted in standard texts such as White and Munson when the loss is referenced to the upstream velocity.

The internal loss of the commercial 4-way splitter body is not published by the vendor. Therefore the code represents that unresolved internal branching loss by a textbook 2-K surrogate for one branch path through the manifold body, optionally scaled by the calibration multiplier `splitterBodyLossMultiplier` (with legacy alias `splitterBodyK`), applied on the connector-velocity basis:

$$\Delta P_{\text{splitter}} = K_{\text{splitter}} \frac{\rho u_c^2}{2}. \tag{228}$$

Accordingly, the modeled distribution-network contribution becomes

$$\Delta P_{\text{dist}} = \Delta P_{\text{header}} + \Delta P_{\text{header}\to\text{splitter}} + \Delta P_{\text{splitter}} + \Delta P_{\text{connector}} + \Delta P_{\text{connector}\to\text{branch}}. \tag{229}$$

For a single electrical group, the total pressure drop reported for a candidate point is the sum of the shared distribution path, the branch-private distribution path, the heater-tube friction plus helical-obstruction terms, the heater-to-aeration expansion, and the aeration-branch pressure requirement.

If multiple electrical series-count groups coexist in parallel, however, the branch flows are not physically constrained to remain equal once the heater-side power, air temperature, density, and helical-obstruction losses differ between groups. In that case the correct reduced-order network statement is not to sum all group drops in series, but to solve one common branch-side pressure difference together with total-flow continuity. The common upstream header path still contributes equally to every branch; the branch-private splitter, connector, heater, expansion, and aeration losses are then evaluated at the redistributed group-specific branch flows until those branch-path drops match. The reported winter total therefore becomes

$$\Delta P_{\text{tot,heat}} = \Delta P_{\text{dist,shared}} + \Delta P_{\text{branch,common}},$$

with

$$\Delta P_{\text{branch},1}(\dot{m}_1) = \Delta P_{\text{branch},2}(\dot{m}_2) = \cdots = \Delta P_{\text{branch,common}}, \qquad \sum_g N_g \dot{m}_g = \dot{m}_{\text{tot}}.$$

If `headerEnabled = false`, then the shared upstream contribution $\Delta P_{\text{dist,shared}}$ vanishes and the modeled distribution loss is entirely branch-private; that is the current active JSON default. Under the current default topology, where all heated branches belong to the same electrical group, this redistributed-flow statement collapses to the same single-group result as the simpler equal-branch-flow treatment.

0.5.6 Two-Node Bed Solve and Duty Calculation

For each candidate operating point, the total heat delivered to the bed is split between the bottom and top nodes using the bottom-fraction parameter f_b:

$$Q_b = f_b Q_{\text{bed,total}}, \qquad Q_t = (1 - f_b) Q_{\text{bed,total}}. \tag{230}$$

The steady two-node bed temperatures are then obtained from

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b \\ T_t \end{bmatrix} = \begin{bmatrix} Q_b + UA_{b,\infty} T_\infty \\ Q_t + UA_{t,\infty} T_\infty \end{bmatrix}. \tag{231}$$

This 2×2 linear system is solved directly at every candidate point. The required full-power duty fraction is then

$$\text{dutyNeeded} = \frac{Q_{\text{req}}}{Q_{\text{bed,total}}}. \tag{232}$$

0.5.7 Sweep Structure and Warm Starts

For the main design sweep, the code loops over pitch, total flow, and voltage. Within a fixed pitch and flow row, the wire-temperature iteration is warm-started from the previous voltage point. This reduces the iteration count because neighboring voltage states tend to have similar mean wire temperatures.

The wire-diameter study uses a targeted subset rather than the entire design grid. For each sampled wire diameter, the code:

1. fixes the comparison-selected pitch;
2. estimates a diameter-scaled voltage center from the comparison-point branch current and the trial-diameter branch resistance at the reference material state, then searches a local voltage window around that center;
3. uses the configured full-range flow subgrid for the wire study rather than a narrow local flow window; and
4. ranks the resulting candidate set with the same lexicographic rule used everywhere else in the study.

This adaptive voltage-centering step is important for physical consistency: when wire diameter changes, branch resistance changes strongly, so a fixed voltage window around the original reference point can miss the relevant operating neighborhood for thick or thin wires.

0.5.8 Interpretation of the Moisture Maps

The moisture figure is therefore a map of model outputs, not independent inputs. At any point on the $(V, \dot{V}_{\mathrm{tot}})$ plane:

1. the imposed total inlet flow determines the inlet mass flow;
2. the heater solution determines the local heated-air temperature;
3. the heated-air density determines the released volumetric flow and perforation exit velocity;
4. the exit velocity and air properties determine the hole-scale convection and mass-transfer coefficients; and
5. the minimum of the diffusion, carrying-capacity, and heat-limited evaporation rates determines the water-loss and latent-load values.

This is why all three moisture-related subplots vary with voltage, even though the horizontal axis is not itself a flow variable.

0.6 Standalone Summer Spot-Cooler / Assist-Blower Mode

The JSON file now also supports a second operating mode, `summer_spot_cooler`. In this mode the Python workflow does *not* call the MATLAB heater export. Instead, it constructs a steady no-heater airflow sweep directly from the same shared wall, aeration-tube, evaporation, and two-node bed equations already documented above. The intent is to estimate how much airflow, spot-cooler duty, assist-blower duty, and substrate water loss are required when the outside air is hot and the aeration system is used as a plenum-fed cooled-air network rather than as an electrical heater. In the current implementation, the Whynter ARC-1030WN sets the plenum supply temperature and the downstream Goorui GHBH 1D7 34 1R5 side-channel blower provides the pressure rise needed to move air through the splitter and aeration tubes.

0.6.1 Summer Boundary Conditions

In the summer mode, ambient air first passes through a spot cooler and enters an insulated plenum. A separate assist blower then draws from that plenum and pushes the cooled air through the aeration network. If the configured target supply temperature is $T_{\text{supply,target}}$, then the requested sensible cooling duty is

$$Q_{\text{spot,req}} = \dot{m}_{\text{air}} c_{p,\text{air}} \max(T_{\infty,\text{summer}} - T_{\text{supply,target}}, 0)\,. \tag{233}$$

If the spot cooler has a configured maximum sensible capacity $Q_{\text{spot,max}}$, the applied unit duty is

$$Q_{\text{spot}} = \min(Q_{\text{spot,req}}, Q_{\text{spot,max}})\,, \tag{234}$$

and the actual supply temperature becomes

$$T_{\text{air,in}} = T_{\infty,\text{summer}} - \frac{Q_{\text{spot}}}{\dot{m}_{\text{air}} c_{p,\text{air}}}. \tag{235}$$

If a coefficient of performance COP is provided, the corresponding spot-cooler electrical power is

$$P_{\text{spot}} = \frac{Q_{\text{spot}}}{\text{COP}}. \tag{236}$$

Thus the inlet air to the aeration network is no longer taken directly from the ambient condition; it is the spot-cooled supply state:

$$T_{\text{air,in}} = T_{\text{supply,actual}}, \tag{237}$$

and the electrical heater is disabled, so

$$P_{\text{tot}} = 0, \qquad I_{\text{tot}} = 0. \tag{238}$$

The assist blower is modeled by a linear fan-curve approximation. If the published rated-duty point is

$$\left(\dot{V}_{\text{rated}}, \Delta P_{\text{rated}}\right), \tag{239}$$

and the published shutoff pressure is

$$\Delta P_{\text{shutoff}}, \tag{240}$$

then the implied free-air flow is

$$\dot{V}_{\text{free}} = \dot{V}_{\text{rated}} \frac{\Delta P_{\text{shutoff}}}{\Delta P_{\text{shutoff}} - \Delta P_{\text{rated}}}. \tag{241}$$

The available blower static at a requested airflow $\dot{V}$ is then

$$\Delta P_{\text{avail}}(\dot{V}) = \max\left[\Delta P_{\text{shutoff}} \left(1 - \frac{\dot{V}}{\dot{V}_{\text{free}}}\right), 0\right]. \tag{242}$$

For the current Goorui GHBH 1D7 34 1R5 basis, the model uses the listing headline values as a temporary two-point fan-curve approximation,

$$\dot{V}_{\text{rated}} = 255\ \text{m}^3/\text{h} \approx 150.1\ \text{CFM}, \qquad \Delta P_{\text{rated}} = 0\ \text{in. H}_2\text{O}, \qquad \Delta P_{\text{shutoff}} = 140\ \text{mbar} \approx 56.2\ \text{in. H}_2\text{O}. \tag{243}$$

The flow sweep is therefore one-dimensional:

$$\dot{V}_{\text{tot}} \in \left[\dot{V}_{\text{min}}, \dot{V}_{\text{max}}\right], \tag{244}$$

with the number of sampled flow points set by the JSON field `cooling_mode.totalFlow_Lpm_count`.

0.6.2 Aeration-Tube Marching Without Electrical Heating

The summer aeration-tube march uses the same segment structure as the heated model, but the heater section is omitted. For each aeration segment, the wall-transfer and jet terms remain

$$q_{\text{wall,seg}} = U_{\text{aer}} P_{\text{out}} \Delta x \left(T_{\text{air}} - T_{\text{bed,ref}}\right), \tag{245}$$

$$q_{\text{jet,seg}} = \dot{m}_{\text{rel,seg}} c_{p,\text{air}} \left(T_{\text{air}} - T_{\text{bed,ref}}\right), \tag{246}$$

while the latent sink is still

$$q_{\text{evap,seg}} = \dot{m}_{\text{evap,seg}} h_{fg}. \tag{247}$$

Accordingly, the net segment contribution to the bed is

$$q_{\text{bed,seg}} = q_{\text{wall,seg}} + q_{\text{jet,seg}} - q_{\text{evap,seg}}. \tag{248}$$

Summing along the tube gives

$$Q_{\text{bed,branch}} = \sum_{\text{seg}} q_{\text{bed,seg}}, \qquad Q_{\text{bed,total}} = N_{\text{tubes}} Q_{\text{bed,branch}}. \tag{249}$$

In summer mode, net cooling corresponds to

$$Q_{\text{bed,total}} < 0. \tag{250}$$

The reported cooling capacity is therefore

$$Q_{\text{cool}} = \max(-Q_{\text{bed,total}}, 0). \tag{251}$$

0.6.3 Required Cooling Load

The same lumped ambient-loss model is reused, but with the summer ambient temperature replacing the winter greenhouse value. If the desired bed reference temperature is $T_{\text{set,cool}}$, the corresponding lumped cooling requirement is

$$Q_{\text{req,cool}} = UA\left(T_{\infty,\text{summer}} - T_{\text{set,cool}}\right). \tag{252}$$

This is the cooling analog of the winter heating-loss requirement and is reported in the standalone summary as a reference capacity target.

0.6.4 Summer Water Requirement

The same segmentwise evaporation model is reused in summer mode, with the inlet relative humidity taken from `cooling_mode.inletRelativeHumidity`. The total evaporated water is

$$\dot{m}_{\text{water,day}} = 86400\, N_{\text{tubes}} \sum_{\text{seg}} \dot{m}_{\text{evap,seg}}. \tag{253}$$

In the current plenum-fed spot-cooler implementation, the reported quantity is the predicted substrate water loss itself rather than a separate external sprinkler-supply multiplier.

0.6.5 Summer Lexicographic Selection Rule

The summer mode also avoids weighted penalties. Instead, each point is ranked lexicographically using the configured order in `cooling_mode.optimization.priority_order`. The default hard-safe residuals are:

$$C_1 = \max(\Delta P - \Delta P_{\text{avail}}(\dot{V}), 0), \tag{254}$$
$$C_2 = \max(\Delta P - \Delta P_{\max}, 0), \tag{255}$$
$$C_3 = \max(\dot{m}_{\text{water,day}} - \dot{m}_{\text{water,max}}, 0), \tag{256}$$
$$C_4 = \max(u_h - u_{h,\max}, 0), \tag{257}$$
$$C_5 = \max(T_b - T_{b,\max}, 0), \tag{258}$$
$$C_6 = \max(T_t - T_{t,\max}, 0), \tag{259}$$
$$C_7 = \max(|T_b - T_t| - \Delta T_{\max}, 0). \tag{260}$$

Among points for which all seven residuals are zero and $Q_{\text{bed,total}} < 0$, the default clean ordering is:

1. minimize $\max(T_b, T_t)$;
2. minimize the mean bed temperature $\frac{1}{2}(T_b + T_t)$;
3. minimize combined cooling-system electrical power, $P_{\text{spot}} + P_{\text{blower}}$;
4. minimize total airflow; and

5. minimize bottom–top spread.

Thus the summer recommendation is based directly on thermal safety, spot-cooler duty, substrate water loss, and blower demand rather than on a weighted penalty score.

0.6.6 Mode Switching and Export Behavior

The JSON runner now uses `run_modes` to control whether the heating branch, the summer-cooling branch, or both are executed in a single call:

1. `run_modes.heating = true`: use the MATLAB-backed voltage-flow heater sweep and write its outputs to `outputs/heating`;
2. `run_modes.cooling = true`: skip the MATLAB heater export and build the airflow-only summer sweep directly in Python, writing its outputs to `outputs/cooling`; and
3. the root file `outputs/study_summary.txt` serves as an index pointing to the mode-specific summaries.

The legacy `operating_mode` field is retained only as a backward-compatible fallback if `run_modes` is not present. This separation is deliberate. The summer mode is physically the same bed, wall, perforation, moisture, and plume model, but it is not a voltage-driven electrical-heater problem, so it should not be represented by forcing the heater sweep to operate at $V = 0$.

0.7 Configuration and JSON-Controlled Inputs

The current workflow uses a shared configuration file, `study_config.json`, to control both the MATLAB export inputs and the standalone Python reporting logic. In the current layout, the main study and output controls appear first, the heater-side export inputs are grouped under `model_overrides` together with the heating `optimization` block, and the cooler-only inputs are grouped later under `cooling_mode`.

0.7.1 Model Overrides That Change the Physics

Edits under `model_overrides` are passed into the MATLAB design model before the export is generated. These settings alter the computed operating points and therefore change the underlying physics results. The most important groups are:

- `model_overrides.sweep`, which defines the voltage, total-flow, pitch, and wire-diameter sweeps, including either explicit arrays or `min/max/count` specifications, as well as the editable report duty target `targetDuty`, now set to 0.75 by default;
- `model_overrides.limits`, which defines explicit limits such as maximum current, wire temperature, outlet-air temperature, pressure drop, and water loss;

- `model_overrides.optimization`, which defines the heating-side thermal feasibility thresholds such as maximum bottom–top spread and minimum bottom and top temperatures;
- `model_overrides.environment`, which exposes the shared heating ambient temperature, ambient pressure, and reference plate length used in the external natural-convection estimate;
- `model_overrides.bin`, which now selects the active vessel-top condition for the export, including the closed-top vented case through `openTop = false`, `ventHoleCount`, and `ventHoleDiameter_m`, while also exposing fill fraction, bulk density, bed heat capacity, effective bed conductivity, and the bottom-node heat-input split;
- `model_overrides.bin_wall`, which contains the editable vessel wall-stack properties such as sheet thickness, sheet conductivity, insulation thickness, insulation conductivity, and the internal bed-to-wall convection coefficient;
- `model_overrides.heater_tube`, which contains the editable base heater-tube dimensions such as tube length, ID, OD, and heater insulation thickness, as well as the heater-wall conductivity, heater-insulation conductivity, external convection coefficient, segment count, and extra heater minor-loss coefficient;
- `model_overrides.heaterTube`, which contains the editable helical-geometry inputs such as `coilMeanD_m` and `coilSpan_m`;
- `model_overrides.aeration`, which includes the effective bed-side sensible coefficient h_{bed}, the perforation release fraction, the active tube count `nParallelTubes`, and the editable aeration-network dimensions such as length, ID, OD, and optional header ID, together with wall conductivity, segment count, the end/header minor-loss coefficients, and the explicit splitter/connector inputs `headerEnabled`, `splitterOutletCount`, `splitterInletID_mm`, `branchConnectorID_mm`, `branchConnectorLength_m`, `headerLossMultiplier`, `splitterBodyLossMultiplier`, and `contractionConeAngle_deg`;
- `model_overrides.perforation`, which includes hole count and hole diameter; and
- `model_overrides.evaporation`, which includes relative humidity, the wetted-area basis, the Table 3 particle-surface inputs, surface temperature, and Lewis factor.

In the current default configuration, `headerEnabled = false`. Hence the model does not assume a separate large upstream header; instead it treats the distribution hardware as a bank of 4-way splitter modules with a nominal 19 mm splitter inlet and branch connector, followed by the larger aeration tube. Contraction losses are evaluated with the Idel'chik Section III, Diagram 3-5 "conical converging bellmouth without end wall" model at the

currently configured angle of 60°. This matches the stated use of the Ground-Work Tractor Supply 4-way brass shutoff manifold as the physical distribution device. Because the vendor page does not publish an internal hydraulic diameter or an internal branch-loss coefficient, the unresolved internal manifold loss is represented by the textbook tee-plus-valve 2-K surrogate, scaled if desired by `splitterBodyLossMultiplier`.

The `summary_inputs` block is intentionally *not* the editable source for these geometry/material quantities. It is used only for static report/specification text when no corresponding physics override is supplied. Accordingly, vessel-wall properties now live under `model_overrides.bin_wall`, base heater-tube dimensions under `model_overrides.heater_tube`, aeration-network dimensions under `model_overrides.aeration`, and coil mean diameter / coil span under `model_overrides.heaterTube`.

If any of these quantities are changed, the export must be regenerated because the candidate operating points themselves will change. The current Python workflow can detect this automatically by hashing the `model_overrides` block and rerunning the MATLAB export when that signature changes.

0.7.2 Standalone Ranking and Reporting Controls

At the top level of the same JSON file, the `optimization` block controls the standalone Python NSGA-II / Pareto heating-selection workflow and the reported-design priority order, while the `radial_profile` block controls the standalone worm-safe-distance plot. The `tube_layout` block controls the standalone cross-sectional tube-placement, spacing, and overlap-aware worm-habitat exclusion-volume calculation. The same file also now contains `run_modes`, which independently enables the MATLAB-backed heater-design branch and the standalone summer airflow-only branch, together with the corresponding `cooling_mode` block. The older `operating_mode` field is retained only as a backward-compatible fallback if `run_modes` is omitted. In particular:

- `method`, `population`, `generations`, and `seed` define the standalone heating NSGA-II settings;
- `objectives` defines the heating objective vector used to build the feasible Pareto set;
- `report_priority_order` defines how one displayed heating design point is selected from that feasible Pareto set;
- `priority_order` is retained for best-available fallback ranking and diagnostics when no feasible Pareto point exists;
- `spread_limit_C`, `min_bottom_temp_C`, and `min_top_temp_C` set the corresponding displayed thermal thresholds; and
- `top_candidate_count` and `n_top` control how many ranked points are printed; and

- `cooling_mode.ambientAir_C`, `cooling_mode.pressure_Pa`, `cooling_mode.totalFlow_Lpm_*`, `cooling_mode.inletRelativeHumidity`, `cooling_mode.surfaceTemp_C`, `cooling_mode.spot_cooler`, `cooling_mode.assist_blower`, and `cooling_mode.optimization` define the standalone summer plenum-fed spot-cooler / assist-blower sweep, the summer hard constraints, and the cooling-mode lexicographic objective; and

- `run_modes.heating` and `run_modes.cooling` determine whether the script writes the corresponding mode-specific folders `outputs/heating` and `outputs/cooling`, while the root file `outputs/study_summary.txt` acts as an index pointing to the two mode-specific summaries; and

- `radial_profile.temperature_threshold_C`, `max_distance_m`, `bed_reference`, `initial_plume_radius_mode`, `spread_rate_m_per_m`, and `max_plume_radius_m` control the post-processed finite-plume radial air-cooling calculation with local re-solved evaporation; and

- `tube_layout.layoutMode`, `tube_layout.maxSupportedTubes`, `tube_layout.wallOffsetWidthFraction`, `tube_layout.wallOffsetHeightFraction`, `tube_layout.includeWallConduction`, `tube_layout.tubeCenters_m`, `tube_layout.tubeCenterHeight_m`, `tube_layout.tubeCenterX_m`, and `tube_layout.cross_section_grid_points` control the cross-sectional tube spacing, the requested rule-based 1-to-12 tube orientation, the optional wall-return correction, and the numerical union-area estimate used to report pipe-occupied, worm-unsafe, and remaining habitable bed volume.

Within `tube_layout`, the precedence is now explicit and consistent: `tubeCenters_m` fully overrides the other placement fields; if that list is empty and `tubeCenterX_m` is provided, the code uses a single-row layout at `tubeCenterHeight_m`; and if `tubeCenterHeight_m` is left null, the code defaults it to the same `wallOffsetHeightFraction` times filled-bed height used by the rule-based bottom row.

Changing only these top-level ranking or wording settings does not require a new MATLAB export if the underlying data file is unchanged.

The current clean ordering is:

1. hard constraints first (current, wire temperature, outlet-air temperature, pressure drop, water loss, minimum bottom temperature, minimum top temperature, and maximum spread);

2. maximize $\min(T_b, T_t)$ among the hard-safe set;

3. minimize total electrical power;

4. minimize $|T_b - T_t|$; and

5. use mean bed temperature and residual heat shortfall only as late tie-breakers.

0.7.3 Practical Interpretation

The JSON structure therefore separates two kinds of edits:

1. **Physics edits:** changes to `model_overrides` modify the actual solved model and require a fresh export.

2. **Post-processing edits:** changes to top-level plot, wording, and ranking settings modify how the current exported dataset is interpreted and displayed.

3. **Summer-mode edits:** changes inside `cooling_mode` alter the standalone summer plenum-fed spot-cooler / assist-blower sweep directly and do not require a MATLAB rerun because no heater-export dataset is used in that mode.

4. **Run-selection edits:** changes to `run_modes` alter only which branches are executed and which output folders are written; they do not alter the underlying physics of either branch.

This distinction is especially important for evaporation. Because the model now uses

$$Q_{\text{bed,branch}} = \sum \left(q_{\text{wall}} + q_{\text{jet}} - q_{\text{evap}} \right), \tag{261}$$

changing evaporation-related JSON inputs is a *physics* change, not a pure reporting change.

0.8 Implementation Reference Map

The MATLAB source comments use the following shorthand labels for the main correlation families and source materials cited throughout this document:

- **[R1]** Churchill, S.W.; Bernstein, M. "A Correlating Equation for Forced Convection from Gases and Liquids to a Circular Cylinder in Crossflow." *Journal of Heat Transfer* 99(2), 1977.

- **[R2]** Gnielinski, V. "New Equations for Heat and Mass Transfer in Turbulent Pipe and Channel Flow." *International Chemical Engineering* 16(2), 1976.

- **[R3]** Churchill, S.W. "Friction-Factor Equation Spans All Fluid-Flow Regimes." *Chemical Engineering* 84(24), 1977.

- **[R4]** Churchill, S.W.; Chu, H.H.S. "Correlating Equations for Laminar and Turbulent Free Convection from a Vertical Plate." *International Journal of Heat and Mass Transfer* 18(11), 1975.

- **[R5]** Bergman, Lavine, Incropera, DeWitt. *Fundamentals of Heat and Mass Transfer*. Used for thermal resistance networks, continuity, energy balances, and lumped-capacitance arguments.

- **[R6]** Idel'chik, I.E. *Handbook of Hydraulic Resistance*, Section IV sudden-expansion loss, written on the upstream-velocity basis. This is the same Borda–Carnot coefficient presented in standard fluid-mechanics texts such as White or Munson.
- **[R7]** Ideal-gas density and Sutherland-law viscosity relations for air.
- **[R8]** Hartenstein, R. "Metabolic parameters of the earthworm *Eisenia foetida* in relation to temperature." *Biotechnology and Bioengineering* 24(8), 1982.
- **[R9]** IAPWS SR1-86(1992). Revised Supplementary Release on Saturation Properties of Ordinary Water Substance.
- **[R10]** Hasan, A.; Siren, K. "The Lewis factor and its influence on the performance prediction of wet-cooling towers." *International Journal of Thermal Sciences* 44(9), 2005.
- **[R11]** Frederickson, J.; Howell, G.; Hobson, A.M. "Effect of pre-composting and vermicomposting on compost characteristics." *European Journal of Soil Biology* 43 (2007) S320–S326. Table 3 supplies the particle-size distribution used in the current wetted-area estimate.

0.9 Year-Round Energy, Cost, and Climate-Driven Operation

This note documents the Python year-round layer that extends the main vermicomposter heating/cooling study. The annual layer is not an independent thermal model. It reuses:

1. the current two-node bed model,
2. the currently selected heating reference point from the MATLAB-backed heating study, and
3. the currently selected cooling reference point from the standalone summer spot-cooler plus assist-blower study.

The new annual layer differs from the earlier monthly `resistiveheatingcode.m` scaffold in one important way: the ambient driver is now a source-backed representative Connecticut climate year rather than a short hard-coded monthly average list plus ad hoc offsets.

1 Climate Data Sources

1.1 Connecticut Monthly Mean and Variability

Monthly ambient statistics are taken from NOAA NCEI Climate at a Glance (CAG), using the statewide Connecticut monthly mean temperature series (`tavg`)

for calendar years 2016–2025. The CAG API is documented at

`https://www.ncei.noaa.gov/monitoring-content/cag/text/time-series-api-doc.html` (262)

and the monthly statewide endpoint pattern used in the study is

`https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/` (263)

`statewide/time-series/6/tavg/1/{month}/2016-2025/data.json` (264)

where state code 6 is Connecticut.

From that NOAA series, the Python model stores for each month m:

$$\bar{T}_m \tag{265}$$

as the 2016–2025 monthly mean temperature in °C, and

$$\sigma_m \tag{266}$$

as the corresponding 2016–2025 monthly standard deviation of that monthly mean series.

1.2 Freeze and Heat-Wave Timing Statistics

Event timing is derived from NOAA daily-summaries station `USW00014740` over the same 2016–2025 period using

`https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries` (267)

with daily `TMAX`, `TMIN`, and `TAVG`.

The annual model uses this daily dataset only to derive event timing and frequency statistics. The ambient baseline itself remains the Connecticut statewide monthly mean series.

2 Representative-Year Construction

2.1 Gaussian-Regression Baseline Profile

The smooth daily baseline is now built by periodic Gaussian kernel regression, not by a sinusoid. Let the month-midpoint day-of-year anchors be

$$x_m, \qquad m = 1, \ldots, 12, \tag{268}$$

with observed monthly mean temperatures

$$\bar{T}_m. \tag{269}$$

For any day of year d, the circular day-distance to month anchor m is

$$\delta(d, x_m) = \min(|d - x_m|,\ 365 - |d - x_m|)\,. \tag{270}$$

With Gaussian kernel bandwidth h, the daily baseline ambient is

$$T_{\text{base}}(d) = \frac{\sum_{m=1}^{12} \exp\bigl(-\delta(d, x_m)^2/(2h^2)\bigr)\, \bar{T}_m}{\sum_{m=1}^{12} \exp(-\delta(d, x_m)^2/(2h^2))}. \tag{271}$$

The same periodic Gaussian regression is also applied to the monthly standard deviation anchors σ_m to build a smooth daily variability envelope, although the primary annual ambient baseline is the regressed mean field $T_{\text{base}}(d)$.

This change was made because the earlier sine-based interpolation imposed an artificial oscillation within every month. The Gaussian-kernel reconstruction is a smoother nonparametric interpolation of the monthly Connecticut climate anchors and is therefore a more defensible baseline for the representative year.

2.2 Freeze-Day Definition and Override

The study uses the operational freeze-day definition

$$T_{\min} \leq 32^\circ\text{F}, \tag{272}$$

applied to the NOAA daily-summaries record.

For each month, the daily-station analysis provides:

1. the average number of freeze days per month,
2. the average day of month on which freeze days occur, and
3. the mean freeze-day anomaly below the corresponding Connecticut monthly mean baseline.

Let the expected freeze-day count for month m be F_m. The code converts the expected monthly counts to a representative integer annual schedule using largest-remainder allocation:

$$n_{f,m} = \text{LR}(F_m), \tag{273}$$

where the total number of allocated freeze days equals the rounded annual expected total.

The freeze days are *not* then forced into one contiguous block. Instead, the code assigns each day-of-month i a Gaussian timing weight centered on the mean observed freeze day $\mu_{f,m}$:

$$w_{f,m}(i) = \exp\left(-\frac{(i - \mu_{f,m})^2}{2s_{f,m}^2}\right), \tag{274}$$

where $s_{f,m}$ is a month-scale spread parameter. Freeze days are then selected sequentially by maximizing a repulsion-modified score

$$\tilde{w}_{f,m}(i) = w_{f,m}(i) \prod_{j \in \mathcal{S}_{f,m}} \left[1 - 0.65 \exp\left(-\frac{(i-j)^2}{2r_{f,m}^2}\right)\right], \tag{275}$$

with $\mathcal{S}_{f,m}$ the set of already selected freeze days and $r_{f,m}$ a repulsion length scale. This preserves the NOAA-derived mean timing but discourages selecting many neighboring days, so the representative freeze days are dispersed through the cold part of the month rather than forming one artificial mid-month block.

If the mean freeze anomaly is $\Delta T_{f,m} > 0$, the ambient profile is overridden by

$$T_m(d) = \min\bigl(T_{\mathrm{base},m}(d), \bar{T}_m - \Delta T_{f,m}\bigr) \tag{276}$$

on the selected freeze days.

2.3 Heat-Wave Definition and Override

The study uses the operational heat-wave definition

$$\text{at least 3 consecutive days with } T_{\max} \geq 90^\circ\mathrm{F}. \tag{277}$$

This threshold is a study definition applied consistently to the NOAA daily record. From the daily station data, the code derives for each month:

1. average heat-wave spell starts per month,
2. average heat-wave days per month,
3. average heat-wave start day of month, and
4. mean heat-wave anomaly above the Connecticut monthly baseline.

If S_m is the expected monthly spell-start count and H_m is the expected monthly heat-wave day count, the code allocates representative integer counts by largest-remainder rounding:

$$n_{s,m} = \mathrm{LR}(S_m), \qquad n_{h,m} = \mathrm{LR}(H_m). \tag{278}$$

Within each month, the spell model enforces the study definition by requiring at least three days per spell. The spell-start candidates are selected from a Gaussian timing distribution centered on the observed mean heat-wave start day $\mu_{h,m}$:

$$w_{h,m}(i) = \exp\left(-\frac{(i-\mu_{h,m})^2}{2s_{h,m}^2}\right), \tag{279}$$

then dispersed by the same repulsion concept used for freeze days:

$$\tilde{w}_{h,m}(i) = w_{h,m}(i) \prod_{j \in \mathcal{S}_{h,m}} \left[1 - 0.65 \exp\left(-\frac{(i-j)^2}{2r_{h,m}^2}\right)\right]. \tag{280}$$

Here $\mathcal{S}_{h,m}$ contains already assigned spell starts and $r_{h,m}$ is scaled to the representative spell spacing. After these starts are chosen, the heat-wave days remain contiguous *within each spell* because that contiguity is part of the heat-wave definition itself. The change is only that the spell starts are now distributed across the month instead of being packed around a single central date.

If the mean heat-wave anomaly is $\Delta T_{h,m} > 0$, the ambient profile is overridden on those heat-wave days by

$$T_m(d) = \max\big(T_m(d), \bar{T}_m + \Delta T_{h,m}\big)\,. \tag{281}$$

2.4 Resulting Representative-Year Ambient Driver

The final ambient driver is therefore a 365-day deterministic sequence constructed from:

1. statewide Connecticut monthly means,
2. statewide monthly variability levels,
3. freeze-day frequency, timing, and cold anomaly from NOAA daily station data, and
4. heat-wave spell frequency, timing, duration, and hot anomaly from NOAA daily station data.

3 Daily Heating and Cooling Loads

Let the current bin conductance to ambient be

$$UA_{\text{bin}}. \tag{282}$$

For each representative day d with ambient temperature $T_\infty(d)$, the required heating load is

$$Q_{\text{req},h}(d) = UA_{\text{bin}} \max(T_{h,\text{set}} - T_\infty(d),\, 0)\,, \tag{283}$$

and the required cooling load is

$$Q_{\text{req},c}(d) = UA_{\text{bin}} \max(T_\infty(d) - T_{c,\text{set}},\, 0)\,. \tag{284}$$

4 Reference-Point Capacity Application

The annual layer no longer treats the selected heating and cooling points as fixed-capacity, fixed-power devices evaluated only at the original design ambients of $-15°$C and $37.5°$C. Instead, it keeps the selected operating *controls* fixed and re-solves those same operating points at each representative day's ambient temperature.

For heating, the fixed controls are the selected branch voltage, total flow, pitch, wire geometry, and current tube/manifold geometry. At each day d with ambient temperature $T_\infty(d)$, the Python annual layer re-solves the same heater-tube and aeration-tube balances used by the design model to obtain an ambient-specific heating capacity and electrical draw:

$$Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d))\,, \tag{285}$$

$$P_{h,\mathrm{elec}}(d) = P_h(T_\infty(d))\,. \tag{286}$$

For cooling, the fixed controls are the selected total flow, spot-cooler settings, and assist-blower configuration. The annual layer then re-solves the standalone cooling point at the day's ambient temperature to obtain

$$Q_{c,\mathrm{cap}}(d) = -Q_{\mathrm{to\,bed},c}(T_\infty(d))\,, \tag{287}$$

with total electrical input

$$P_{c,\mathrm{elec}}(d) = P_{\mathrm{spot\,cooler}}(d) + P_{\mathrm{assist\,blower}}(d). \tag{288}$$

For each day, the required duty fractions are

$$D_h^*(d) = \frac{Q_{\mathrm{req},h}(d)}{Q_{h,\mathrm{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\mathrm{req},c}(d)}{Q_{c,\mathrm{cap}}(d)}, \tag{289}$$

and the applied duties are clipped at unity:

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1). \tag{290}$$

If the day is a heating day, the applied bed load is

$$Q_{\mathrm{bed}}(d) = D_h(d)\, Q_{h,\mathrm{cap}}(d), \tag{291}$$

whereas on a cooling day it is

$$Q_{\mathrm{bed}}(d) = -D_c(d)\, Q_{c,\mathrm{cap}}(d). \tag{292}$$

5 Daily Bottom and Top Bed Temperatures

The current two-node steady model is then solved each day using the same bed split fraction f_b as the main study:

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b(d) \\ T_t(d) \end{bmatrix} = \begin{bmatrix} f_b Q_{\mathrm{bed}}(d) + UA_{b,\infty} T_\infty(d) \\ (1-f_b) Q_{\mathrm{bed}}(d) + UA_{t,\infty} T_\infty(d) \end{bmatrix}. \tag{293}$$

Therefore the annual temperature plot is not heuristic. It is the same bed energy-balance model already used in the heating and cooling studies, solved day by day under the representative Connecticut climate driver.

6 Daily Electricity, Cost, and Water

Daily electricity use is

$$E_h(d) = \frac{P_{h,\mathrm{elec}}(d)\, D_h(d) \cdot 24}{1000}, \qquad E_c(d) = \frac{P_{c,\mathrm{elec}}(d)\, D_c(d) \cdot 24}{1000}, \tag{294}$$

in kWh/day. Total daily electricity use is

$$E(d) = E_h(d) + E_c(d). \tag{295}$$

This matters because the annual electricity model now reflects the fact that heater-shell losses, air density, wire resistance, spot-cooler sensible load, and assist-blower duty all vary with the day's ambient condition. The annual cost is therefore no longer based on simply scaling the single design-point power values found at $-15°$C and $37.5°$C.

If the Connecticut electricity price is c_{elec} in USD/kWh, then daily operating cost is

$$C(d) = c_{\text{elec}} E(d). \tag{296}$$

The current default JSON value is

$$c_{\text{elec}} = 0.2875 \text{ USD/kWh}, \tag{297}$$

based on the U.S. EIA Electric Power Annual 2024, Table 2.10, Connecticut residential average retail electricity price.

The daily water replacement estimate scales the already-computed heating or cooling water-loss rate:

$$m_{\text{water}}(d) = \dot{m}_{\text{water},24h}\, D(d), \tag{298}$$

so the annual layer does not create a second, inconsistent moisture model.

7 Monthly Aggregation

After the daily solution is built, the code aggregates by month to report:

1. ambient minimum, maximum, mean, and source-backed monthly standard deviation,
2. freeze-day and heat-wave counts,
3. mean and worst-case bottom/top bed temperatures,
4. mean and peak daily duty,
5. monthly heating and cooling electricity,
6. monthly operating cost, and
7. monthly water replacement.

The time-series figure written to `outputs/year_round/year_round_energy_cost_analysis.png` contains:

1. ambient, bottom-node, and top-node temperatures versus day of year,
2. explicit freeze-day and heat-wave day markers, and
3. daily plus cumulative operating cost versus day of year.

8 Relation to `resistiveheatingcode.m`

The local MATLAB file `resistiveheatingcode.m` was used as the starting point for the annual idea, but the current Python model is more physically consistent because it:

1. removes the older ad hoc monthly solar and passive-cooling offsets,
2. replaces the short hard-coded monthly temperature list with NOAA-backed Connecticut climate inputs, and
3. computes daily bottom and top bed temperatures from the same current two-node model used in the rest of the study.

9 Model Credibility, Scope, and Sign-Off Requirements

9.1 Physics Blocks with the Strongest Current Basis

For the present design workflow, the following blocks have the strongest engineering basis and are appropriate for comparative design decisions:

1. temperature-dependent Nichrome electrical resistance and resistive heat generation in the wire;
2. series–parallel electrical loading of identical heater branches;
3. forced convection and friction-factor-based pressure drop in the heater and aeration tubes using established internal-flow correlations;
4. heater-shell loss through the tube wall and insulation by a standard series-resistance formulation;
5. perforation continuity, hole exit velocity, and the bookkeeping of useful bed heating as
$$Q_{\text{to bed}} = \sum \left(q_{\text{wall}} + q_{\text{jet}} - q_{\text{evap}} \right); \tag{299}$$
6. natural-convection vessel loss to greenhouse air using established vertical- and horizontal-surface correlations; and
7. year-round re-solving of the already-selected heating and cooling reference points under a representative climate driver rather than assuming fixed design-point power throughout the year.

These blocks should be regarded as the most reliable parts of the current model for ranking candidate voltages, flows, wire diameters, vessel-top configurations, and electrical topologies.

9.2 Reduced-Order or Screening-Level Blocks

Other blocks are still physically motivated, but they remain reduced-order or screening approximations:

1. the two-node bed model, which resolves only a bottom and top substrate temperature rather than a full spatial temperature field;
2. the covered-top-with-small-vent extension, which treats the vent as an area-limited top opening and does not explicitly resolve buoyancy-driven exchange through the vent jet;
3. the splitter/manifold network model, for which the unresolved internal losses are represented by a lumped splitter coefficient;
4. the evaporation model, which is internally consistent and source-backed but still depends on a simplified wetted-area estimate and a Lewis-relation mass-transfer closure;
5. the radial warm-air plume and worm-habitat exclusion calculation, which is a post-processing estimate rather than a resolved flow field; and
6. the summer spot-cooler / assist-blower representation, which is appropriate for feasibility screening but still relies on a simplified fan curve, simplified spot-cooler performance mapping, and a reduced psychrometric closure.

Accordingly, the present model is best interpreted as a design and optimization tool rather than a certification-grade hardware prediction.

9.3 What the Current Model Is Appropriate For

In its present form, the model is appropriate for:

1. comparing candidate heater geometries, flow rates, wire diameters, and electrical topologies;
2. identifying hard-constraint-safe operating regions for current, wire temperature, outlet-air temperature, pressure drop, water loss, and bed temperature thresholds;
3. ranking vessel-top configurations such as uncovered, covered, and covered with a small vent hole;
4. estimating seasonal electricity, water, and cost trends; and
5. selecting the most defensible prototype configurations for laboratory or greenhouse testing.

It should not, by itself, be treated as sufficient evidence for final hardware release, guaranteed worm-safe local exposure, or formal thermal safety sign-off under all faulted operating conditions.

9.4 Requirements for Final Hardware Sign-Off

A final hardware sign-off model would require all of the following:

1. a frozen as-built geometry, including the actual heater wire route, support details, electrical joints, clearances, manifold geometry, vent geometry, and enclosure details;

2. measured electrical and thermal properties for the purchased hardware, including the actual wire resistance–temperature relation, insulation performance, contact resistances, and shell temperatures;

3. measured hydraulic data for the real distribution hardware, especially the blower curve, splitter/manifold loss coefficients, connector losses, and exit losses, rather than placeholder or lumped estimates;

4. transient validation tests at representative winter and summer boundary conditions, including startup, steady operation, duty-cycled operation, and shutdown;

5. explicit fault-case evaluation, including blocked or reduced airflow, blower failure, vent blockage, relay failure, and sensor failure;

6. uncertainty quantification showing that the validated model error band is small enough that all limits are still met with margin after uncertainty is applied; and

7. a higher-fidelity substrate model capable of resolving local hot spots, temperature gradients, and moisture migration in the bed.

Only after those requirements are met should the model be treated as a final hardware sign-off basis rather than as a prototype-selection tool.

9.5 What a 2D/3D Conjugate Porous/Moist Model Would Entail

For this project, a sign-off-grade higher-fidelity model would be a *conjugate* model because it would solve the coupled physics in the solid hardware, the flowing air, and the porous moist substrate simultaneously rather than treating the bed as only two lumped temperature nodes.

At minimum, such a model would include:

1. conduction in the metal and insulated hardware domains, including the heater tube, aeration tubes, vessel walls, and cover;

2. fluid flow in the internal air passages and, if resolved, through the perforations and the nearby free-plume region;

3. porous-medium flow and heat transfer in the substrate, with air exchange between the perforations and the bed;

4. transient energy storage in the substrate and walls;
5. moisture transport in the substrate, including evaporation, humidification of the gas phase, and the associated latent heat sink; and
6. realistic boundary conditions for greenhouse air, cover openings, and all forced-flow inlets and outlets.

In practical terms, this means solving coupled conservation equations for:

1. mass and momentum in the open-flow regions;
2. energy in the wire, tube walls, insulation, air, and porous substrate;
3. species transport for water vapor in the gas phase; and
4. liquid-moisture storage and migration in the porous substrate, either by a saturation-based porous-media model or by an experimentally calibrated effective-moisture formulation.

Relative to the current model, the main benefit would be the ability to predict spatially resolved quantities such as:

1. local worm-exposure temperatures near perforations and near the tube surface;
2. local bed drying or wetting zones;
3. nonuniform temperature fields caused by tube placement, wall proximity, and vent placement; and
4. transient recovery and overshoot during startup, shutdown, and disturbed airflow events.

The main cost would be substantially higher model complexity, substantially greater computational time, and a much larger experimental burden for calibration and validation. For that reason, such a model is usually reserved for final design verification after the reduced-order model has already narrowed the design space to one or two promising hardware configurations.

10 Sources

1. NOAA NCEI Climate at a Glance API documentation: `https://www.ncei.noaa.gov/monitoring-cont`
2. NOAA NCEI Climate at a Glance Connecticut statewide monthly mean temperature endpoint pattern: `https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glan`
3. NOAA daily-summaries endpoint used for event statistics: `https://www.ncei.noaa.gov/access/servic`
4. U.S. Energy Information Administration, Electric Power Annual 2024, Table 2.10: `https://www.eia.gov/electricity/annual/html/epa_02_10.html`

5. The main study heat-transfer and moisture equations documented in `VermiHeatLoss_complete_updated.` and `vermi_water_optimization_numerics_appendix.tex`; the annual layer reuses that same two-node bed model and the currently selected heating and cooling reference points.